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Review article

Modern statistical models for forensic fingerprint examinations: A critical review

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ABSTRACT

Over the last decade, the development of statistical models in support of forensic fingerprint identification has been the subject of increasing research attention, spurned on recently by commentators who claim that the scientific basis for fingerprint identification has not been adequately demonstrated. Such models are increasingly seen as useful tools in support of the fingerprint identification process within or in addition to the ACE-V framework.

This paper provides a critical review of recent statistical models from both a practical and theoretical perspective. This includes analysis of models of two different methodologies: Probability of Random Correspondence (PRC) models that focus on calculating probabilities of the occurrence of fingerprint configurations for a given population, and Likelihood Ratio (LR) models which use analysis of corresponding features of fingerprints to derive a likelihood value representing the evidential weighting for a potential source.

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1. Introduction

Fingerprints have been widely used throughout the world as a means of identification for forensic purposes. Forensic experts have extensively relied on the premises that fingerprint characteristics are highly discriminatory and immutable amongst the general population. For the majority of the 20th century, the forensic identification of fingerprints has had near unanimous acceptance as robust forensic evidence, where testimonies provided by fingerprint experts were rarely challenged and the philosophical foundations of such testimonies were rarely questioned. However, in recent times, there has been a number of questions raised regarding the scientific validity of forensic fingerprint identification [1–4].

The current wave of scrutiny in North America is largely associated with the Daubert decision [5] by the Supreme Court in the USA, concerning expert evidence admissibility. In the 1993 case of Daubert v. Merrell Dow Pharmaceuticals [6] a ruling was made that outlined criteria concerning the admissibility of scientific expert testimony, based somewhat on criteria used in the broader scientific community. The criteria for a valid scientific method were stated as being as follows:

- must be based on testable and falsifiable theories or techniques,
- must be subjected to peer-review and publication,
- must have known or predictable error rates,
- must have standards and controls concerning its applications, and
- must be generally accepted by a relevant scientific community.

The guidelines for expert testimony admissibility from Daubert have since influenced international jurisdictions, with the UK Law Commission's recent expert evidence consultation paper [7] also prescribing similar standards (with an emphasis on scientific method more so than falsifiability).

Fingerprint identification via the ACE-V (Analysis, Comparison, Evaluation, and Verification) framework [9] has been subjected to scrutiny from a number of academics and legal commentators, who have cited the following objections [8]:

- the unfounded and unfalsifiable theoretical foundations of fingerprint feature discriminability,
- the 'unscientific' absolute conclusions of identification in testimonies (i.e., either identification, exclusion, or inconclusive), and
- the contextual bias of experts for decisions made within the ACE-V framework.

While contextual bias is primarily concerned with influences on accuracy and consistency of practitioners within the ACE-V process [10–12], the remaining criticisms can be restated as the non-existence of a scientifically sound probabilistic framework for fingerprint evidential assessment, that has the consensual approval from the forensic science community.

The traditional theoretical foundations of fingerprint identification primarily rest on observational science, where a high discriminability of feature characteristics exists. However, there is a lack of consensus regarding quantifiable error rates for a given pair of 'corresponding' feature configurations [13]. Some critics have invoked a more traditional interpretation for discriminability [14,15], claiming that an assumption of 'uniqueness' is used. This clearly violates the falsifiable requirement of Daubert. However, more and more experts do not necessarily associate discriminability with uniqueness [16]. Nevertheless, a consensus framework for calculating accurate error rates for corresponding fingerprint features needs to be established.

The conclusions of identification [20] made by fingerprint practitioners have historical influence from Edmond Locard's *tripartite rule* [21]. The tripartite rule is as follows:

- Positive identifications are possible when there are more than 12 minutiae within sharp quality fingermarks.
- If 8–12 minutiae are involved, then the case is borderline. Certainty of identity will depend on additional information such as finger mark quality, rarity of pattern, presence of the core, delta(s), and pores, and ridge shape characteristics, along with agreement by at least 2 practitioners.
- If a limited number of minutiae are present, the fingermarks cannot provide certainty for an identification, but only a presumption of strength proportional to the number of minutiae.

In a holistic sense, the tripartite rule can be viewed as a probabilistic framework, where the successful applications of the first and second rules are analogous to a statement with certainty that the mark and the print share the same source, whereas the third rule covers the probability range between 0% and 100%. While some jurisdictions only apply the first rule to set a numerical standard within the ACE-V framework, other jurisdictions (such as Australia, UK, and USA [22]) adopt a holistic approach, where no strict numerical standard or feature combination is prescribed. Nevertheless, current fingerprint practitioner testimony is largely restricted to conclusions that convey a statement of certainty, ignoring the third rule's probabilistic outcome.

A probabilistic framework for fingerprint identification has not been historically popular and was even previously banned by professional bodies (see [23]). In recent times however, a probabilistic framework for fingerprint identification has had more favourable treatment within the forensic community. It was suggested in [24] that a probabilistic framework is based on strong scientific principles unlike the traditional numerical standards. In

support of this, the IAI have recently rescinded their ban on reporting possible, probable, or likely conclusions (see [25]) and support the future use of valid statistical models (provided that they are accepted as valid by the scientific community) to aid the practitioner in identification assessments. In addition, the European Fingerprint Working Group (EFPWG) of the European Network of Forensic Science Institutes (ENFSI) has also advocate for the use of statistical models [26]. Research into the use of statistical models for fingerprint identification purposes is ongoing and still in its infancy in terms of operational deployment. Currently, no known models have certification or official endorsement from any forensic identification bodies for use in casework. However, recent developments with key forensic bodies, such as the IAI, provide further rationale for model development and pave the way for the use of statistical models within the ACE-V framework.

Given this evolving situation, it is timely to offer a critical review of the models published after the seminal review by Stoney [27] and the chapter he published in 2001 [28]. It is aimed at providing an up-to-date assessment of the various research initiatives that were devoted to the statistical analysis of fingerprints. The paper is focused on the work carried out on minutiae and concentrates on published papers in peer-reviewed journal. Research work presented in conferences (without peer-reviewed proceedings) have not been integrated into the present review.

2. Foundations of statistical models for fingerprint identification

Statistical models for fingerprint identification provide an essential scientific framework in order to quantify the discriminability of given fingerprint feature configurations and to predict associated error rates. Such models can be largely classified as either Probability of Random Correspondence (PRC) models or Likelihood Ratio (LR) models, although there are relationships between the two.

2.1. Foundations of Probability of Random Correspondence models

The primary task of Probability of Random Correspondence (PRC) models is to calculate probability measures concerning the occurrence of corresponding features within impressions sourced from *different* fingerprints. From a fingerprint practitioner's perspective, these probability measures are akin to calculating the probability of close non-match occurrences for feature configurations found within a given fingermark and exemplar(s). Since the very first model proposed by Galton [29] in the late 1800s, there have been over 20 models proposed in the peer reviewed literature, with a rich variety of methodologies used to model the various statistical relationships of fingerprint features that exists.

Generally speaking, PRC models are used to calculate at least one the following probability measures, given a feature set I with m features:

- $PRC(m)$: the probability that two feature configurations from fingermarks of different fingers (randomly selected) have at least m corresponding landmark features (e.g., minutiae, pores, or ridge shape details),
- $PRC(I)$: the probability that a specific feature configuration has a given subset of features, I, corresponding to a subset of features from a randomly chosen configuration from another fingermark from a different finger.
- $nPRC(m)$: the probability that from a set of n feature configurations from fingermarks of different fingers, at least one pair (from a total of $\binom{n}{2}$ possible pairs) have at least m corresponding features. Note that when $n = 2$, $nPRC(m) = PRC(m)$.
- $nPRC(I)$: the probability that a specific feature configuration from a set of n (sourced from fingermarks of different fingers), has a given subset of features, I, corresponding to at least one other subset of features from different configuration(s).
- $EPIC(A, B, m)$: given m observed corresponding features between two different fingerprints A and B, the *Evidence of a Paired Impostor Correspondence* (EPIC) is defined as the probability of obtaining at least m corresponding *synthetic* features randomly generated from respective individualised fingerprint feature models, f_A and f_B. The probability distribution for the number of matching synthetic features is derived from comparing realisations of f_A and f_B. Unlike the $PRC(m)$ calculation, $EPIC(A, B, m)$ performs a conditional analysis by restricting the population to only include A and B.

All PRC probability metrics are derived from some form of model representation for features, either generalised for an entire population or individualised for each fingerprint. Both $PRC(m)$ and $nPRC(m)$ measures can include topographical information, such as finger type, pattern classification or ridge counts between well defined global landmarks. A basic example of a PRC calculation can be found in [30], where each corresponding minutiae, core-to-delta ridge counts, and the matching pattern classification of two randomly chosen fingerprints were assigned a probability of $\frac{1}{4}$, altogether giving

$$PRC(m) = \underbrace{\left(\frac{1}{4}\right)^{m}}_{\text{minutiae}} \cdot \underbrace{\left(\frac{1}{4}\right)}_{\text{core-to-delta r.c}} \cdot \underbrace{\left(\frac{1}{4}\right)}_{\text{class}} = \left(\frac{1}{4}\right)^{m+2} \quad (1)$$

for m minutiae. While historical PRC models largely focus simply on the $PRC(m)$ metric, the two $nPRC$ variants and $EPIC$ are recent developments which have a direct relationship to Automatic Fingerprint Identification System (AFIS) environments.

2.2. Foundations of Likelihood Ratio models

A *likelihood ratio* (LR) is a simple yet powerful statistic when applied to a variety of forensic science applications, including inference of identity of source for evidence such as DNA [31,32], ear prints [33], glass fragments [34], speaker recognition [35] and fingerprints [36,38–41]. An LR is defined as the ratio of two likelihoods of a specific *event* occurring, each of which follow a different hypothesis, and thus, empirical distribution. In the forensic identification context dealing with impressions, an event, E, may represent the recovered mark in question, while the hypotheses considered for calculating the two likelihoods of E occurring are:

- H_P: E comes from a specific known source, P, and
- H_D: E has an alternative source than P.

The LR can be expressed as

$$LR = \frac{P(E|H_P)}{P(E|H_D)} \quad (2)$$

where $P(E|H_P)$ is the likelihood of the observations on the mark and print given that the mark was produced by the same finger as the print P, while $P(E|H_D)$ is the likelihood of the observations on the mark and print given that the mark was *not* produced by the same donor as P. The LR value can be interpreted as follows:

- $LR < 1$: the observations provide more support for hypothesis H_D,
- $LR = 1$: the observations provide equal support from both hypotheses, and
- $LR > 1$: the observations provide more support for hypothesis H_P.

The general LR form of Eq. (2) can be restated specifically for fingerprint identification evaluations. Given an unknown query impression, x (e.g., unknown mark), with m marked features

(denoted as $x^{(m)}$), and a known impression, y (e.g., known AFIS candidate exemplar), with m' marked features (denoted as $y^{(m')}$), the LR is defined as

$$LR = \frac{P(x^{(m)}|y^{(m')}, H_P)}{P(x^{(m)}|y^{(m')}, H_D)} \tag{3}$$

where the value $P(x^{(m)}|y^{(m')}, H_P)$ represents the probability that mark x is observed if x and y were produced by the same finger, while $P(x^{(m)}|y^{(m')}, H_D)$ is the probability that impressions x and y were *not* produced by the same finger, using the *closest* $m \leq m'$ corresponding features between $x^{(m)}$ and $y^{(m')}$. Thus, hypotheses used to calculate the LR numerator and denominator probabilities are defined as:

- H_P: x and y were produced by the same finger, and
- H_D: x and y were produced by different fingers.

The LR of Eq. (3) can be reformulated [37] to give an LR values for the identification of individuals with

$$LR_{pers} = \frac{\sum_{g=1}^{10} P(x^{(m)}|y_g^{(m_{yg})}, H_P) \cdot P(G=g|I_{cs})}{(1/N)\sum_{i=1}^{N}\sum_{g=1}^{10} P(x^{(m)}|y_{i,g}^{(m_{yi,g})}, H_P) \cdot P(G=g|I_{cs})} \tag{4}$$

where g is the finger index, N is the number of prints in a reference set to consider by incorporating ten-prints probabilities, and I_{cs} denotes observations made with regard to how mark x was applied.

In order to derive the within-finger and between-finger probabilities of a feature configuration given by the numerator and denominator of Eq. (3), respectively, some models use the similarity score distributions produced by AFISs [41–43] as a proxy for direct assessment. For a given AFIS similarity score, $s \in S$, in a discrete sample space S, the LR is simply calculated as

$$LR = \frac{P(s|H_P)}{P(s|H_D)}. \tag{5}$$

The numerator and denominator probabilities can be further marginalised for a particular configuration size (m),

$$LR = \frac{P(s|m, H_P)}{P(s|m, H_D)}, \tag{6}$$

or configuration ($y^{(m')}$)

$$LR = \frac{P(s|y^{(m')}, H_P)}{P(s|y^{(m')}, H_D)}. \tag{7}$$

While scores that are found in a continuous sample space can be transformed to a discrete space, an alternative calculation for Eq. (5) is formulated (without discretising scores) as

$$LR = \frac{P(s' \leq s|H_P)}{P(s' \geq s|H_D)}. \tag{8}$$

Alternatively, other models [38,39] make use of statistical distributions of a dissimilarity measure, $d(\bullet, \bullet)$, defined on *Feature Vectors* (FVs) that represent minutiae configurations. Such distributions are used to derive LR values:

$$LR = \frac{P(d(x_d, y_d)|H_P)}{P(d(x_d, y_d)|H_D)} \tag{9}$$

where x_d and y_d are FVs for equally sized minutiae sub-configurations $x^{(m)}$ and $y^{(m')}$, respectively.

The AFIS and FV based LR models re-interpret the original LR hypotheses of Eq. (3) by using score and dissimilarity metric distributions in order to reduce the dimensionality of the evidence at hand. Some have argued that such models are merely statistical propositions that may not necessarily be strictly presentable for legal purposes, specifically when statistical expertise is lacking [44]. Nonetheless, such statistical propositions are valuable tools for the assessment of fingerprint evidence.

2.3. Relationship between PRC and LR models

A clear theoretical relationship exists between PRC and LR models. Since the probabilities calculated from a PRC model solely focuses on corresponding features from impressions with different fingerprint sources, the $PRC(m)$ measure can be expressed simply as the denominator of the LR in Eqs. (5) and (6), with

$$PRC(m) = P(s|m, H_D) \tag{10}$$

where all compared configurations have exactly m features to consider, or more generally, a *PRC* agnostic of specific configuration size

$$PRC = \int_{m_0}^{\infty} PRC(m) = P(s|H_D), \tag{11}$$

where m_0 is the minimum number of minutiae that the AFIS matching algorithm requires to assess correspondence. In addition, for any given dataset containing n exemplars sourced from fingers other than x, the weighted average of the denominator of Eq. (3) converges almost surely to the $PRC(I)$ measure:

$$\left[\frac{1}{n}\sum_{j=1}^{n} P(x^{(m)}|y_j^{(m_j)}, H_D)\right] \xrightarrow{a.s.} PRC(I \equiv x^{(m)}) \tag{12}$$

with $m \leq m_j$, as $n \to \infty$ with *almost sure* convergence by Strong's law of large numbers. Similar expressions can also be found for the $nPRC(I)$ and $nPRC(m)$ measures.

In comparison to the LR calculation, we can clearly see that the PRC value focuses only on assessing the probability of corresponding evidential features arising from fingermarks of different fingers. Thus, PRC models lack the important evidential consideration of within-finger feature variability. However, being a probability value (with range [0, 1]), the interpretation of the PRC measures is sometimes considered as more intuitive than the LR (with range $[0, \infty)$).

An alternative probabilistic metric, called the Non-Match Probability (NMP) (see [43]), can be used to further illustrate the theoretical relationship that exists between both PRC and LR measures. The NMP is the probability that a known print, y, and an unknown fingermark, x, do *not* come from the same source finger. Thus, a higher NMP implies a stronger case for the exclusion of y. With an AFIS context, the NMP can be written mathematically as

$$NMP = P(H_D|s) = 1 - P(H_P|s), \tag{13}$$

which is simply the complement of the probability that the H_P hypothesis is true, given prior conditions of an AFIS score, s.

The NMP of Eq. (13) can be re-written to be expressed in terms of LR and PRC measures. Given prior knowledge of the probabilities $P(H_P)$ and $P(H_D)$, which represent H_P and H_D, respectively, as the ground truth for a given fingerprint assessment, we can apply *Bayes* formula to get the equivalent NMP expression

$$NMP = \frac{P(s|H_D).P(H_D)}{P(s|H_D).P(H_D) + P(s|H_P).P(H_P)}. \quad (14)$$

The above equation clearly shows the dependencies between NMP and the prior probabilities $P(H_D)$ and $P(H_P)$, respectively, while both $P(s|H_P)$ and $P(s|H_D)$ have been introduced in the LR measures. While prior probabilities $P(H_D)$ and $P(H_P)$ are not easily calculated, they can each be assigned a value 0.5 when specifically searching for cold hits. Since the denominator of Eq. (14) can be written as

$$P(s|H_D).P(H_D) + P(s|H_P).P(H_P) = P(s), \quad (15)$$

which is the probability that the features of x and y show a certain degree of correspondence, the NMP can be rewritten as a direct expression of the $PRC(m)$ and LR measures, with

$$NMP = \frac{P(s|H_D) \cdot P(H_D)}{P(s)} = \frac{PRC \times P(H_D)}{P(s)} \quad (16)$$

and

$$NMP = \frac{1}{1 + (P(s|H_P) \cdot P(H_P))/(P(s|H_D) \cdot P(H_D))} = \frac{1}{1 + LR \times (P(H_P)/P(H_D))}, \quad (17)$$

respectively. A simple algebraic relationship is now evident between the measures PRC and LR by equating the right hand sides of Eqs. (16) and (17).

3. Modern PRC models

3.1. Contrasting Modern and historic PRC models

A rich collection of PRC models have recently been developed, all of which use significantly different statistical methods from historical counterparts. The motivation for new model development stems from the following pitfalls that historical models largely suffer from:

- assumption driven: strong assumptions are made regarding the probabilities of feature events without supportive empirical evidence,
- simplistic statistical modelling: features are usually modelled as independent events, with little or no considerations given for inter-feature statistical relationships,
- scarcity of data: AFIS systems were largely unavailable for aiding the development of these models
- no evaluation framework: no evaluation of model fit, accuracy, and associated error rates are considered.

Modern PRC models are AFIS-centric, which provides both a rich dataset of impressions and the automation of within and between-finger feature searching. Such a technological advantage has been used to aid a *data-driven* statistical framework from which feature characteristics are learnt and model accuracy can be evaluated. In addition, more complex or hidden statistical relationship between features has been considered in modern models without the need to make unsubstantiated assumptions of feature traits probabilities, such as that given in Eq. (1).

Unlike historic models, most modern PRC models construct feature models for each fingerprint, representing the fingerprint population in a more detailed and individualised manner. Features are usually randomly generated from these feature models to help calculate PRC values. In addition, these synthetic features are used to assess how well these feature models represent key characteristics of real features, through statistical goodness-of-fit tests or comparisons of AFIS impostor (i.e., between finger) distributions for real and synthetic features.

Modern PRC models are largely based on various generative probability modelling methods. This includes such statistical methods as point processes which are used to model spatial patterns of minutiae, mixture models for clustering spatial neighbourhoods and modelling minutiae location-orientation dependencies, Bayesian networks or hierarchical frameworks to combine feature sub-models, and MCMC simulations or EM algorithm to estimate unknown parameters.

3.2. Spatial homogeneity probability models

Spatial homogeneity models are built on the assumption of the spatial distribution of minutiae having a uniform density throughout all regions of a fingerprint. In addition, such models assume the same density of minutiae for all fingerprints (using the empirically observed average density). While the authors of these models recognise that this does not encompass the entirety of minutia spatial characteristics, they suggest that using such spatial assumptions help derive a straight forward model that produces conservative estimates for PRC values (i.e., higher PRC values in favour of the defendant).

3.2.1. Pankanti et al. [45]

A proposition for a spatial homogeneity model can be found in [45]. This was one of the first attempts to create an AFIS centric model primarily based on rudimentary principals of minutiae matching algorithms. The following assumptions were used in the development of the model:

- Most of the discriminatory power of AFIS are based on minutiae location, type (bifurcations and ridge endings), and direction.
- Minutiae pairing events are independent and of equal importance.
- Modelling of minutiae spatial location and direction can be adequately achieved by using uniform and independent distributions.
- Ridge width is assumed to be uniform across the entire population. This assumption allows the model to ignore local ridge frequency variations and ridge count information between minutiae.
- Fingerprint image quality is not taken into account due to the subjectiveness of quality ratings. Thus, it is assumed all minutiae detected are true and of good quality.
- There exists only one correct alignment between input (I) and matching template (T) fingerprints. Matching minutiae $\{x_i, y_i, \theta_i\}$ and $\{x'_j, y'_j, \theta'_j\}$ from template and input fingerprints, respectively, meet the following distance criterion:

$$sdist_r(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\}) = \sqrt{(x_i - x'_j)^2 + (y_i - y'_j)^2} \leq r_0, \quad (18)$$

along with the direction criterion:

$$dist_\theta(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\}) = \min(|\theta_i - \theta'_j|, 360^\circ - |\theta_i - \theta'_j|) \leq \theta_0. \quad (19)$$

The correct minutiae pairings represent a 1-1 mapping which is analogous to their being only one correct alignment.

Given the total area of overlap, A, between I and T once the correct alignment of features has taken place (see Fig. 1), minutiae pairings between minutiae in I and T can be constructed if a potential pairing has distance and orientation differences within the defined tolerances, r_0 and θ_0, respectively. Since an assumption of independence was made for minutiae location and direction, the probabilities of a minutiae pairing being within the location and orientation tolerances is

$$P(dist_r(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\}) \leq r_0) = \frac{\text{area of tolerance}}{\text{total area of overlap}} = \frac{\pi r_0^2}{A} = \frac{C}{A} \tag{20}$$

and

$$P(dist_\theta(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\})) \leq \theta_0) = \frac{\text{angle of tolerance}}{\text{total angle}} = \frac{2\theta_0}{360^\circ}. \tag{21}$$

Ignoring angle constraints for the moment, while assuming I and T contain n and m minutiae, respectively, the probability of matching p minutiae and not matching $n - p$ minutiae between I and T is

$$P(pairs = p|A, C, m, n) = \underbrace{\left(\frac{mC}{A}\right) \cdot \left(\frac{(m-1)C}{A-C}\right) \cdots \left(\frac{(m-p-1)C}{A-(p-1)C}\right)}_{p \text{ terms}} \times \underbrace{\left(\frac{A-mC}{A-pC}\right) \cdots \left(\frac{A-(m-(n-p+1))C}{A-(n-1)C}\right)}_{n-p \text{ terms}} \tag{22}$$

where the remaining area of tolerance for each minutiae pairing and unpaired minutiae is taken into account. Letting $M = A/C$ and assuming that M is an integer, Eq. (22) reduces to the hyper-geometric distribution:

$$P(pairs = p|A, C, m, n) = \frac{\binom{m}{p} \cdot \binom{M-m}{n-p}}{\binom{M}{n}}. \tag{23}$$

Given the probability of $q \leq p$ out of the p pairs meet the directional criterion as:

$$\binom{p}{q} (l)^q \cdot (1-l)^{p-q} \tag{24}$$

where $l = (2 \times \theta_0/360^\circ)$ is the probability of two spatially matched minutiae having similar direction, the PRC (which is the probability of matching q minutiae in *both* direction and spatial configuration) is the hyper-geometric/binomial mixture model:

$$PRC(q) = \sum_{p=q}^{\min(m,n)} \underbrace{\frac{\binom{m}{p} \cdot \binom{M-m}{n-p}}{\binom{M}{n}}}_{\text{spatial}} \cdot \underbrace{\binom{p}{q} (l)^q \cdot (1-l)^{p-q}}_{\text{directional}}. \tag{25}$$

Lastly, the authors considered that minutiae can only lie on ridges and that ridges occupy roughly $A/2$ of the overlap area. Given an average global ridge period of w, the value $M = A/C$ was change to $M = (A/w)/(2r_0)$ where $2r_0$ is the length of tolerance in minutiae location, while A/w is the total ridge length.

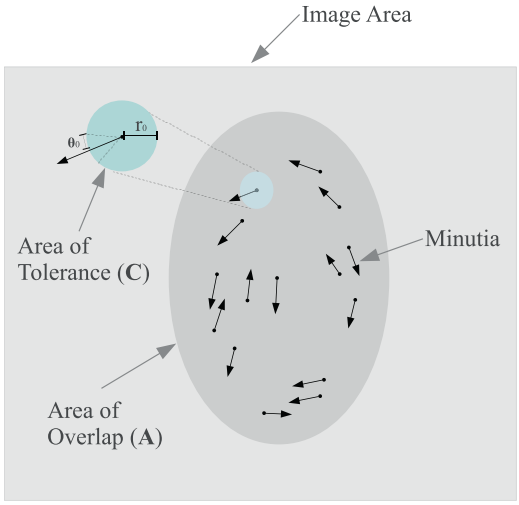


Fig. 1. Minutia pair spatial and directional match tolerance illustrated on a given fingerprint region.
Image adapted from [45].

The parameters, r_0, θ_0, l, A, m, n were estimated using a few different databases with ground truth information. Tuning for the parameters r_0 and θ_0 were achieved by finding:

$$P(dist_r(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\}) \leq r_0) \geq 0.975 \tag{26}$$

and

$$P(dist_\theta(\{x_i, y_i, \theta_i\}, \{x'_j, y'_j, \theta'_j\})) \leq \theta_0) \geq 0.975, \tag{27}$$

respectively, accounting for 97.5% of variability for each marginal distribution.

For the ridge frequency parameter, w, results already obtained in [46] that formulated an estimate for the ridge frequency to be 0.463 mm/ridge, were utilised to give $w \approx 9.1$, after taking into account the resolution (500 dpi) of the template images. Another two databases (each containing 2672 images) were used to evaluate model parameters A, m, and n, along with performing an experimental evaluation of the model. Some of the theoretical PRC values are given in Table 1.

Experimentation compared two distributions of the proposed PRC of Eq. (25) (each tuned on two different databases) against respective empirical impostor minutiae pairing count distributions, as discovered by the fingerprint matching algorithm found in [47]. Results indicated that the distributions from the theoretical model did not fit the empirical distributions well. In particular, the probability for larger numbers of impostor minutiae pairings per an impostor comparison was understated by the theoretical distributions. The authors suggested that reasons for this were that the matching algorithm attempts to maximise correspondences of

Table 1
Sample correspondence probability calculations for the model in [45].

M, m, n, q	PRC
70, 12, 12, 8	6.19×10^{-10}
70, 12, 12, 9	4.88×10^{-12}
70, 12, 12, 10	1.96×10^{-14}
70, 12, 12, 11	3.21×10^{-17}
70, 12, 12, 12	1.22×10^{-20}

minutiae, regardless if the resulting alignment is plausible or not. Also, since minutiae feature extraction was fully automated, spurious minutiae may exist in some of the recorded minutiae pairings, resulting in an inflated number of impostor minutiae pairings.

3.2.2. Chen et al. [48]

The authors of [48] proposed a model loosely based on the assumptions and combinatorial methodology of the model in [45], where minutiae pairing criteria (defined earlier in Eqs. (18) and (19)) were used as a core component of the model development. However, minutiae pairing directional differences of impostors were modelled and assumed not be uniform. In addition, the spatial patterns of minutiae are assumed to have complete spatial randomness (CSR). This was modelled using a homogeneous Poisson point process, in order to generate simulated minutiae, from which PRC estimates were calculated.

The directional differences of impostor minutiae pairings discovered from AFIS were modelled using a tuned von-Mises distribution:

$$P_\theta(\theta_d) = \frac{\exp(\kappa \cdot \cos(2\theta_d - \pi))}{2\pi \cdot I_0(\kappa)} \tag{28}$$

where θ_d is the angular difference, while the function $I_0(\bullet)$ is a modified Bessel function of the first kind with order 0 and $\kappa = 1.69$.

Prior to applying the minutiae pairing directional difference distribution to the model, spatial information was solely considered by the authors. Assuming that all minutiae pairs, (p_i, q_j), adhere to the restriction of having their Euclidean distance less than r_0 after alignment, the probability of spatial random correspondence is $\pi r_0^2/A$. The probability of p_i (from a template set, T) not matching q_j (from a query set, Q) spatially is $C = 1 - \pi r_0^2/A$. Hence, the probability that p_i can be matched spatially to at least one of n query minutiae is $1 - C^n$. This was applied to the generic case of matching at least ρ minutiae pairs from n and m minutiae (from Q and T, respectively):

$$P_S(m,n,\rho) = (1-C^n)P_S(m-1,n-1,\rho-1) + C^n P_S(m-1,n,\rho) \tag{29}$$

which is a first-order linear homogeneous difference equation with three variables, with initial conditions defined as

$$P_S(m,n,\rho) = 0, \quad (m < \rho \text{ or } n < \rho)$$

$$P_S(m,n,\rho) = C^{mn}, \quad (\rho = 0). \tag{30}$$

Through mathematical induction, this spatial probability measure can be expressed as

$$P_S(m,n,\rho) = \frac{C^{(m-\rho)(n-\rho)} \prod_{i=0}^{\rho-1}((1-C^{m-i})(1-C^{n-i}))}{\prod_{i=1}^{\rho}(1-C^i)}. \tag{31}$$

In considering both minutiae location and direction, it was assumed that matching minutiae pairings have directional differences less than or equal to some value, θ_0. If a given minutia, $q_j \in Q$, is spatially corresponding to k minutiae from the template, T, after alignment, then the probability that there exists at least one directional match within the k was given as

$$P_D(n,k) = 2\gamma \times \left(1 - \frac{1}{2^k}\right) \tag{32}$$

where

$$\gamma = \int_0^{\theta_0} P_\theta(x)\,dx \tag{33}$$

with 2γ accounting for orientational difference (not directional), while a probability of 1/2 was assigned for minutiae directional agreement.

Thus, a *single* minutia from Q having k possible neighbouring matching minutiae from T both spatially and directionally after alignment is

$$\eta(n,k,\gamma) = \underbrace{\binom{n}{k} \cdot (1-C)^k \cdot C^{n-k}}_{\text{spatial}} \cdot \underbrace{2\gamma \times \left(1 - \frac{1}{2^k}\right)}_{\text{directional}}, \tag{34}$$

and for at least one matching minutiae:

$$\eta(n,\gamma) = \sum_{k=1}^{n} \eta(n,k,\gamma). \tag{35}$$

Using the conditional probability of a matching minutiae pairing occurring given spatially agreement, as

$$\tau(n,\gamma) = \frac{\eta(n,\gamma)}{1-C^n}, \tag{36}$$

a conservative probability considering both spatial and directional agreement for at least q minutiae pair matches was given as:

$$P_{match}(m,n,q) = \sum_{\rho=q}^{\min(m,n)} \left[P_S(m,n,\rho) \cdot \binom{\rho}{q} \cdot \tau(n,\gamma)^q (1-\tau(n,\gamma))^{(\rho-q)} \right]. \tag{37}$$

Given the CSR assumption for minutiae spatiality, the theoretical distribution for the number of impostor minutiae pairings was simulated using the PRC calculation:

$$PRC(q) = \sum_{m=q}^{\infty} \sum_{n=q}^{\infty} (poiss(m,\lambda_0) \cdot poiss(n,\lambda_0) \cdot P_{match}(m,n,q)) \tag{38}$$

where $poiss(\bullet)$ is the probability function of the Poisson distribution and λ_0 is the average fingerprint minutiae density. An experiment was performed on selected fingerprints from three fingerprint databases where the empirical impostor distribution was compared to the simulated distribution of the proposed model, in addition to the model found in [45]. The experimental results indicated that the simulated distribution of the proposed method was much closer to the observed empirical distributions.

3.2.3. Model methodology analysis

The proposed models attempt to replicate results from AFIS algorithms, mirroring what most modern day practitioners would deal with everyday. However, some deficiencies exist in the assumptions made for the models. The assumption of independence for minutiae spatial and directional detail in [45] was noted as going against experimental evidence, as shown originally in [46,49]. This assumption was, however, made purely to provide an upper estimate of a random correspondence occurring. In other words, a significant overestimation of this occurrence is given, as the dependence of minutiae spatial location and orientation is strong (specially in areas away from singularities due to the slow curvature of fingerprint patterns). Ultimately, this approach is

conservative in a prosecution situation. Although [48] excludes assuming a uniform distribution for the directional differences of spatially agreeing minutiae, their assumption of a homogeneous minutiae spatial distribution strongly goes against empirical observations [59], ultimately leading to inaccurate PRC estimates.

3.3. Spatio-directional based generative models

Spatio-directional based generative models attempt to model general minutiae location and direction dependencies using a family of finite continuous distribution based mixture models. Unlike spatial homogeneity models, both minutia spatial/directional clustering tendencies and dependencies were modelled. However, any inter-minutiae dependencies are ignored and minutiae are treated as independent and identically distributed random events. The PRC value is derived from feature models created per fingerprint configuration in a given dataset.

3.3.1. Dass et al. [50]

The authors of [50] proposed the first spatio-directional generative model. Given a random minutia location and direction, denoted as $\mathbf{s}_j = (x_j, y_j) \in \mathbb{R}^2$ and $\theta_j \in [0, 2\pi)$, respectively, a r.v. (random vector) is defined for each minutia in a k-minutiae configuration as $\mathbf{x}_j = (\mathbf{s}_j, \theta_j)$ for $j = 1, 2, \ldots, k$. In order to model the dependence between $\mathbf{s}_j$ and θ_j, the joint mixture density was defined as:

$$f(\mathbf{s}_j, \theta_j | \Theta_G) = \sum_{g=1}^{G} \tau_g \cdot \underbrace{f_X(\mathbf{s}_j | \mu_g, \Sigma_g)}_{\text{spatial}} \cdot \underbrace{f_D \theta_j | \nu_g, \kappa_g, \rho_g)}_{\text{directional}}, \quad (39)$$

where G is the number of components (i.e., representing minutiae location/orientation clusters) in the mixture model with corresponding weights, τ_g, $f_X(\mathbf{s}_j|\mu_g, \Sigma_g)$ is the spatial component modelled by the probability density function of a bivariate Gaussian r.v. with mean μ_g and covariance matrix Σ_g, $f_D(\theta_j|\nu_g, \kappa_g, \rho_g)$ is the directional component defined by

$$f_D(\theta_j|\nu_g, \kappa_g, \rho_g) = \rho_g \upsilon(\theta) \cdot I\{0 \le \theta < \pi\} + (1 - \rho_g)\upsilon(\theta - \pi) \cdot I\{\pi \le \theta < 2\pi\}, \quad (40)$$

with p_g as the probability of a minutia having direction θ (where $\theta \in [0, \pi)$) as opposed to $\theta + \pi$ (with probability $1 - p_g$), $I\{A\}$ is the indicator function of the set A, and $\upsilon(\theta)$ is the Von-Mises distribution

$$\upsilon(\theta_j) \equiv \upsilon(\theta_j|\nu_g, \kappa_g) = \frac{2}{I_0(\kappa_g)} \exp\{\kappa_g \cos 2(\theta_j - \nu_g)\} \in [0, \pi) \quad (41)$$

with ν_g and κ_g as the mean angle and precision, respectively. The set of unknown parameters, Θ_G, consists of G and $(\mu_g, \Sigma_g, \nu_g, \kappa_g, \tau_g)$ for $g = 1, 2, \ldots, G$. The number of components, G, was calculated using Bayes Information Criteria (BIC), whereas the EM algorithm was used to estimate each unknown parameter $(\mu_g, \Sigma_g, \nu_g, \kappa_g, \tau_g)$. Example of this model fitted on a fingerprint is illustrated in Fig. 2.

A PRC calculation was proposed using the minutiae spatial/directional feature models. Given two fingerprints Q and T, the probability that a specific number of minutiae pairings between Q and T was calculated as

$$p(m|Q,T) = \binom{n}{m} \cdot (p(1|Q,T))^m \cdot (1 - p(1|Q,T))^{n-m} \quad (42)$$

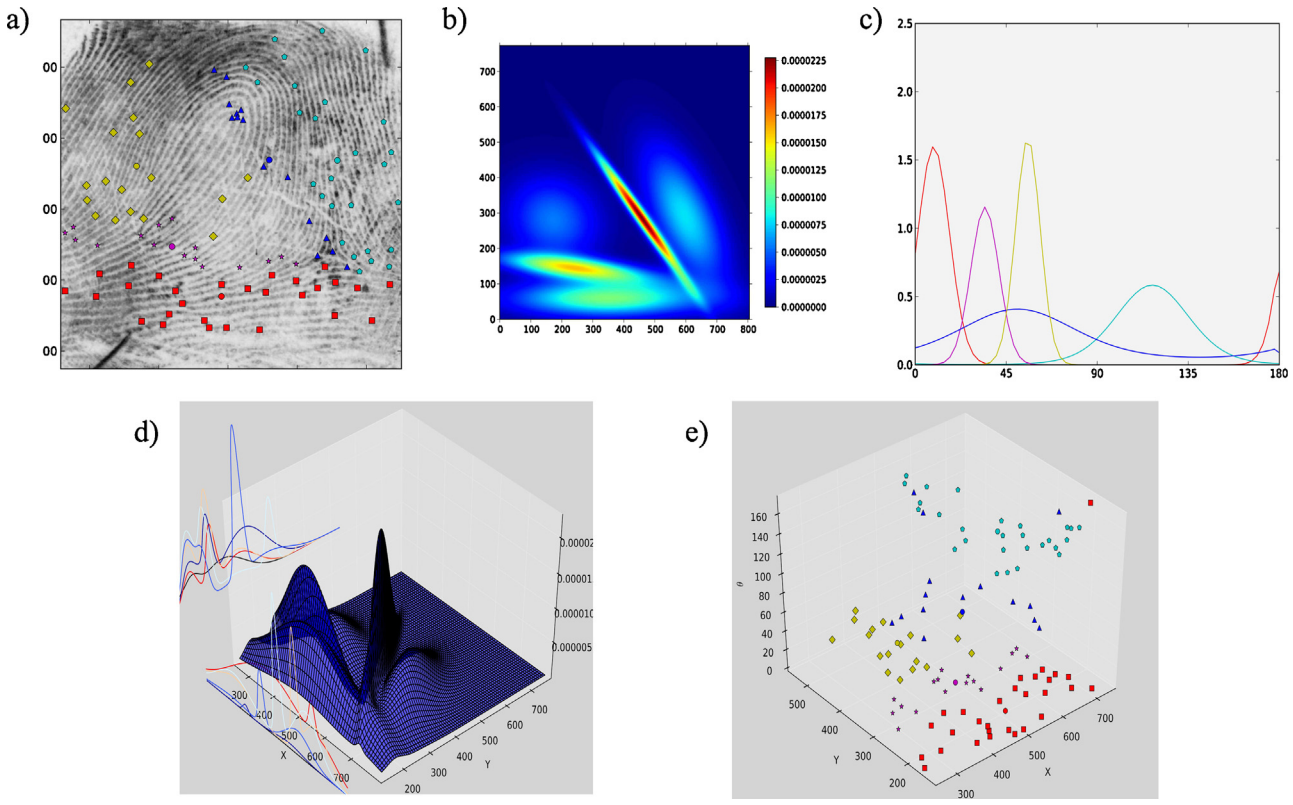


Fig. 2. (a) A NIST4 [18] fingerprint with categorised minutiae clusters derived from the mixture model with parameters tuned by the EM algorithm. (b) Spatial density map of mixture model. (c) Directional density of mixture model clusters. (d) Three dimensional view of the spatial density map. (e) Three dimensional view of spatial-directional clusters.

where m is the number of matched synthetic minutiae between Q and T, and $p(1, Q, T)$ is the probability for one of the n minutiae from Q matching one of the n' of T (assuming $n \leq n'$). Since most matching algorithms produce at least one matching minutiae pair, the probability, $p(1, Q, T)$, was estimated by calculating the value of the conditional expectation (given at least one match) as:

$$m_0 = \frac{n \cdot p(1|Q,T)}{(1-(1-p(1|Q,T))^n)} \tag{43}$$

where m_0 can be directly estimated by simulating minutiae features from the proposed minutiae model for both Q and T, followed by determining the number of observed minutiae matches using the previously prescribed matching algorithm. An estimate of $p(1|Q, T)$ was extracted via numerical methods. Noting that a fingerprint database containing L impressions of N fingers will have $(N-1)L$ impostor comparisons for each configuration, then a total of $N(N-1)L^2$ impostor comparisons exist in a database. Thus, *PRC* is given by

$$PRC(m) = \frac{1}{N(N-1)L^2} \sum_{(Q,T)\in I} p(m|Q,T) \tag{44}$$

where I is the set of all possible impostor fingerprint pairings.

As per the previous minutiae location/direction model analysis, a comparative experiment of the PRC was set up with the same dataset, along with the matching algorithm in [51]. The assessment in this experiment was between the proposed *PRC* value against the calculation described in [45]. The proposed *PRC* calculations were illustrated to be more closer to empirical PRC values discovered from the matching algorithm.

3.3.2. Zhu et al. [52]

The model proposed in [52] was largely based on the model found in [50], where each fingerprint had its features modelled stochastically. However, modifications to the PRC calculation were made and similar feature models were grouped together via a clustering technique.

For a given query fingerprint Q with n minutiae and a tuned minutiae feature model denoted as $f_Q = f(\mathbf{s}, \theta|\Theta_G^Q)$ (i.e., Eq. (39)), and likewise, template fingerprint with n' minutiae and tuned minutiae model denoted as $f_T = f(\mathbf{s}, \theta|\Theta_G^T)$, synthetic minutiae ($\mathbf{s}_i$, θ_i) and ($\mathbf{s}_j$, θ_j) randomly generated from feature models, f_Q and f_T, respectively, have the matching pair probability:

$$p(1|Q,T) \equiv p((\mathbf{s}_i,\theta_i),(\mathbf{s}_j,\theta_j)) = P(|\mathbf{s}_i - \mathbf{s}_j|_s \leq r_0 \text{ and } |\theta_i - \theta_j|_a \leq \theta_0) \tag{45}$$

where

$$|\mathbf{s}_i - \mathbf{s}_j|_s \equiv \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{46}$$

and

$$|\theta_i - \theta_j|_a \equiv \min(|\theta_i - \theta_j|, 2\pi - |\theta_i - \theta_j|). \tag{47}$$

Using the Poisson distribution function, the probability of finding exactly m minutiae pairs was given as:

$$p(m|Q,T) = \frac{e^{-\lambda(Q,T)}\lambda(Q,T)^m}{m!}. \tag{48}$$

where $\lambda(Q, T)$ is the expected rate of matching minutiae between Q and T, defined as

$$\lambda(Q,T) = (n \cdot n') \cdot p(1|Q,T), \tag{49}$$

which is the total number of possible pairings $(n \cdot n')$ multiplied by the probability for a matching minutiae pair. The proposed PRC was then calculated in a similar manner as in [50], with

$$PRC(m) = \frac{2}{F(F-1)} \sum_{(Q,T)\in I} p(m|Q,T) \tag{50}$$

where I is the set of all impostor fingerprint pairings and $2/F(F-1)$ is the inverse of the total number of impostor matches (i.e., acting as a normalisation constant). Feature models were built from *super templates*, which combine distinct minutiae found from all impressions of a particular finger into a single template, while model fit assessment was performed using the Chi-square and Freeman–Tukey goodness-of-fit tests. A more robust PRC calculation using the *α-trimmed mean*, was proposed:

$$PRC_\alpha(m) = \frac{2}{F(F-1)(1-\alpha)} \sum_{(Q,T)\in I} p_\alpha(m|Q,T). \tag{51}$$

where $p_\alpha(m|Q, T)$ is adjusted to represent the $1-\alpha$ confidence interval of $p(m|Q, T)$. This was to remove outlier minutiae simulations which may skew the PRC value.

Further modifications were made to feature models, where an agglomerative hierarchical clustering procedure was used on the space of all fitted fingerprint mixture models. A dissimilarity measure between two mixture models f_Q and f_T was given by the Hellinger distance:

$$H(f,g) = \int_{s\in\mathbb{R}^2}\int_{\theta\in[0,360^\circ)} \left(\sqrt{f_Q(s,\theta)} - \sqrt{f_T(s,\theta)}\right)^2 ds d\theta \tag{52}$$

where $0 \leq H(f_Q, f_T) \leq 2$ with $H(f_Q, f_T) = 0$ if and only if $f_Q = f_T$. For a given database., there are $F(F-1)/2$ Hellinger distances for each impostor comparison in which the clustering will be applied. Defining the N clusters of mixture densities as C_1, C_2, …, C_N, a threshold for the Hellinger distance was given as t, where $t = 2$ will give $N = 1$ and $t = 0$ will give N as $\lceil F(F-1)/2 \rceil$. The *within* cluster dissimilarity is defined as

$$W_N = \sum_{i=1}^{N} \frac{1}{2|C_i|} D(C_i) \tag{53}$$

where

$$D(C_i) = \sum_{f_Q, f_T \in C_i} H(f_Q, f_T) \tag{54}$$

is the sum of all distance $H(f_Q, f_T)$ for models, f_Q and f_T, in cluster C_i, and $|C_i|$ is the size of cluster C_i. The selection for number of clusters, N, was chosen using the *elbow criteria*, where by defining $G_N = |W_N - W_{N-1}|$, all $N' > N$ have $W_{N'} \approx 0$.

The per cluster mean density feature models replaced the original features models used, modifying the PRC calculation. Upon choosing a value for N, the mean density for each cluster C_i is defined as

$$\overline{f}(\mathbf{s},\theta) = \frac{1}{|C_i|} \sum_{Q\in C_i} f_Q(\mathbf{s},\theta). \tag{55}$$

Using the selected value of N, the probability of matching exactly m minutiae between Q and T (Eq. (48)) was restated as

$$p(m|Q,T) = \frac{e^{-\lambda(C_Q,C_T)}\lambda(C_Q,C_T)^m}{m!} \tag{56}$$

where Q and T belong to clusters C_Q and C_T, respectively.

3.3.3. Su et al. [53]

The model proposed in [53] extended the minutiae feature model to account for ridge curvature information. Ridges are represented as a set of ridge points sampled at equal intervals of inter ridge width from each detected minutiae with polar coordinates (r_i, ϕ_i) about the source minutia. Using the maximum encountered ridge length in the FVC2002 database [17], L, three ridge lengths were defined as

- short ridges: $l_r \le L/3$ with no sampling performed,
- medium ridges: $L/3 < l_r < 2L/3$ with $L/3$ points sampled,
- long ridges: $2L/3 \le l_r \le L$ with both $L/3$ and $2L/3$ points sampled.

The mixture density function of Eq. (57) incorporating both ridge and minutia direction/location features, $\mathbf{x}_m$, was proposed,

$$p(\mathbf{x}_m|\Theta) = \begin{cases} p^l(l_r).\sum_g^{G1}\pi_g\, p_g(\mathbf{s}_m,\theta_m|\Theta_g) & \text{if } l_r \le L/3, \\ p^l(l_r).\sum_g^{G2}\pi_g\, p_g(\mathbf{s}_m,\theta_m|\Theta_g).\, p_g^{\lfloor L/3\rfloor}(r_{\lfloor L/3\rfloor}\phi_{\lfloor L/3\rfloor},\theta_{\lfloor L/3\rfloor}|\Theta_g) & \text{if } L/3 < l_r < 2L/3, \\ p^l(l_r).\sum_g^{G3}\pi_g\, p_g(\mathbf{s}_m,\theta_m|\Theta_g).\, p_g^{\lfloor L/3\rfloor}(r_{\lfloor L/3\rfloor},\phi_{\lfloor L/3\rfloor},\theta_{\lfloor L/3\rfloor}|\Theta_g) & \\ \cdot\, p_g^{\lfloor 2L/3\rfloor}(r_{\lfloor 2L/3\rfloor},\phi_{\lfloor 2L/3\rfloor},\theta_{\lfloor 2L/3\rfloor}|\Theta_g) & \text{if } 2L/3 \le l_r \le L, \end{cases} \tag{57}$$

where $p^l(a, b)$ is the uniform distribution on the interval $[a, b]$, $G1$, $G2$, $G3$ are the number of mixture model components for each ridge length category, π_g are the component weights, $p_g^{\lfloor L/3\rfloor}$ and $p_g^{\lfloor 2L/3\rfloor}$ are the ridge density for the $\lfloor L/3\rfloor$th and $\lfloor 2L/3\rfloor$th sample points, respectively, and $p_g(\mathbf{s}_m, \theta_m|\Theta_g)$ is the minutia density for location s_m and direction θ_m (i.e., mixture component distribution used in Eq. (39)). Each ridge density probability function p_g^i at the ith ridge point is defined by direction and location components

$$p_g^i(r_i,\phi_i,\theta_i|\Theta_g) = \mathcal{N}(r_i|\mu_{ig},\sigma_{ig}).V(\phi_i|\nu_{ig}^{\phi},\kappa_{ig}^{\phi},\rho_{ig}^{\phi}).V(\theta_i|\nu_{ig}^{\theta},\kappa_{ig}^{\theta},\rho_{ig}^{\theta}) \tag{58}$$

where $\mathcal{N}(r_i|\mu_{ig}.\sigma_{ig})$ is a univariate Gaussian density function multiplied with Von-Mises density functions representing ridge point and minutiae directional distributions. Unknown parameters were estimated using the EM algorithm with component sizes validated via k-means clustering results (best k-means clustering results chosen).

The PRC calculation was reworked from the approaches described in [50,52]. The probability of a random minutia and associated ridge points, $\mathbf{x}_a$, matching another minutia/ridge feature combo, $\mathbf{x}_b$, is

$$\begin{aligned} p_\epsilon(\mathbf{x}) &= p(|\mathbf{x}_a - \mathbf{x}_b| \le \epsilon|\Theta) \\ &= \int_{\mathbf{x}_a}\int_{|\mathbf{x}_a-\mathbf{x}_b|\le\epsilon} p(\mathbf{x}_a|\Theta).\, p(\mathbf{x}_b|\Theta)d\mathbf{x}_a d\mathbf{x}_b \end{aligned} \tag{59}$$

where $\epsilon = \{(\epsilon_s, \epsilon_\theta), (\epsilon_r, \epsilon_\phi)\}$ are the tolerances defined for both minutiae and associated ridge points, and Θ is the set of parameters describing minutiae and ridge distributions. The PRC of matching *at least* m pairs of minutiae from two configurations is defined as

$$p_\epsilon(m|m_1,m_2) = \underbrace{\binom{m_1}{m}\binom{m_2}{m}m!}_{\#\text{ of config. corresp.}} \times \underbrace{p_\epsilon(\mathbf{x})^m(1-p_\epsilon(\mathbf{x}))^{(m_1-m)(m_2-m)}}_{\text{match/non-match probabilities}} \tag{60}$$

where m_1 and m_2 are the respective minutiae counts of each configuration.

The PRC derived from a set of n fingerprints is calculated as

$$PRC(n) = 1 - (1 - p_{match})^{(n(n-1))/2} \tag{61}$$

where p_{match} is the probability of matching two fingerprints from n, given by

$$p_{match} = \sum_{m_1\in M_1}\sum_{m_2\in M_2} p_c(m_1)\cdot p_c(m_2)\cdot p_\epsilon(m|m_1,m_2), \tag{62}$$

where M_1 and M_2 contain the minutiae counts of the n fingerprints, and $p_c(C)$ is the minutiae count probability per configuration.

Given a specific fingerprint configuration f, the *nPRC* is calculated as

$$nPRC(f,m) = 1 - (1 - p_{f_\epsilon}(m))^{n-1} \tag{63}$$

where $p_{f_\epsilon}(m)$ is the probability that m minutiae/ridge combo pairs match from fingerprint f with a randomly chosen fingerprint from a set of n,

$$p_{f_\epsilon}(m) = \sum_{m_1\in M} p_c(m_1)\binom{m_1}{m}\cdot\sum_{i=1}^{\binom{m_f}{m}}\prod_{j=1}^{m} p(\mathbf{x}_{ij}|\Theta) \tag{64}$$

where m_f is the number of minutiae in f, and M is the set of possible minutiae tally values.

An experiment was performed using the FVC2002 DB1 database [17]. Table 2 contrasts the different PRC values for calculating with both minutiae and ridge information against only minutiae. The PRC values are clearly smaller when ridge detail is taken into account, as there are more features available to distinguish configurations.

3.3.4. Other related models

The core feature model of Eq. (39) was extended in [54] to include ridge shape information associated with each minutia. Ridge detail, r, attached to each minutia, $(\mathbf{s}, \theta)$, was classified as one of the sixteen possible ridge shapes. The empirical distribution of

Table 2
Ridge and non-ridge PRC values from the FVC2002 DB1 database.

Configuration	PRC with ridge	PRC w/o ridge
$m=16, n=16, w=4$	1.6×10^{-3}	2.1×10^{-1}
$m=16, n=16, w=8$	1.7×10^{-8}	7.8×10^{-3}
$m=16, n=16, w=16$	3.1×10^{-24}	1.6×10^{-11}
$m=26, n=26, w=6$	7.9×10^{-4}	1.4×10^{-1}
$m=26, n=26, w=12$	3.8×10^{-10}	5.4×10^{-4}
$m=26, n=26, w=20$	2.4×10^{-22}	5.4×10^{-11}
$m=26, n=26, w=26$	1.2×10^{-35}	2.1×10^{-20}
$m=36, n=36, w=6$	4.1×10^{-3}	1.7×10^{-1}
$m=36, n=36, w=12$	8.5×10^{-13}	2.8×10^{-5}
$m=36, n=36, w=20$	1.6×10^{-27}	4.2×10^{-14}
$m=36, n=36, w=26$	3.6×10^{-49}	7.3×10^{-30}

classified ridge shapes (for each cluster g) with density function, $f_R^g(r)$, was used to extend the feature model to

$$f(\mathbf{s},\theta,r|\Theta_G)=\sum_{g=1}^{G}\tau_g\cdot f_X(\mathbf{s}|\mu_g,\Sigma_g)\cdot f_D(\theta|\nu_g,\kappa_g)\cdot f_R^g(r), \tag{65}$$

Another extension can be found in [55], where ridge period and curvature associated with each minutia in the gth cluster was added to the feature model definition:

$$f(\mathbf{s},\theta,r,c|\Theta_G)=\sum_{g=1}^{G}\tau_g\times\underbrace{f_X(\mathbf{s}|\mu_g,\Sigma_g)\cdot f_D(\theta|\nu_g,\kappa_g)}_{\text{minutiae spatial/direction}}\times\underbrace{f_R(r|\omega_g,\sigma_g^2)\cdot f_C(c|\lambda_g)}_{\text{ridge period/curvature}}, \tag{66}$$

where ridge period and curvature were modelled using Gaussian and Poisson distributions, respectively. This was further extended to include core spatial detail of ridges associated with each minutiae.

3.3.5. Model methodology analysis

One shortcoming in the spatio-directional based models is that only macro scale spatial characteristics are discovered, while any statistical spatial relationships within clusters (i.e., between neighbouring minutiae) are ignored. In addition, the resulting spatio-directional clusters from the EM algorithm with BIC model selection can exhibit poor fit. For example, Fig. 2(c) illustrates a high variance for the orientation representation of the some groups, while the spatial distribution of minutiae within clusters can over-disperse rather than focus on the mean spatial location of the cluster.

Another shortcoming can be found in the model fit analysis methods used to assess the fit of the feature models, all of which suffer from some theoretical deficiencies. For instance, the Ripley's K was function used in [50], which is a spatial point process test that focuses on dispersion measure of random point samples (testing the fit of spatial detail only). In addition, the binned goodness-of-fit statistical tests which were used (Chi-square and Freeman–Tukey) to test the fit of the clusters in [52] have varying fit assessment results depending on chosen bin size configurations and have stronger assumptions than robust non-parametric techniques such as [56].

In real world applications involving fingerprints retrieved from crime scenes, ridge detail is largely noisy and incomplete. Thus, practical applications, the models of [55,54] will often need to degenerate to the minutiae only models found in [50,52].

3.4. Bayesian networks based generative model

All of the modern models that have been discussed thus far do not measure any inter-minutiae dependencies that may arise in a given population, treating all minutiae as independent and identically distributed events. In support of using inter-minutiae dependency in a model, it was found in [46] that spatially close minutiae tend to have the same direction and that the variance of minutia direction at a particular fingerprint region is dependent on the spatial variance of the minutiae. Also, minutiae spatially distribute differently on different test scales. On a relatively small scale, minutiae tend to over-disperse, while clustering occurs when observed at a larger scale. In order to capture the distribution of minutiae along with any dependencies between them, a generative model based on Bayesian networks was proposed in [57]. Bayesian networks are a powerful statistical modelling technique which represent a set of random variables and their conditional dependencies (i.e., statistical relationships) using directed acyclic graphs. Bayesian networks are used widely in forensic science, particularly for DNA profiling [58].

3.4.1. Su et al. [57]

The model proposed in [57] extends the spatio-directional feature model found in [50] to include neighbouring inter-minutiae dependencies using Bayesian networks based on a defined *minutiae sequence*. A minutiae sequence is defined starting with the closest minutia to the core, $\mathbf{x}_1$, followed iteratively by the next minutia, $\mathbf{x}_n$, which is the closest minutia to the spatial arithmetic mean of the previous $n-1$ minutiae. From this, a sequence $\mathbf{X}=\{\mathbf{x}_1,\ldots,\mathbf{x}_n\}$ is defined for all minutiae (see Fig. 3 for an example with 5 minutiae). From the constructed minutiae sequence, the joint distribution was defined as

$$p(\mathbf{X})=p(\mathbf{s}_1)\,p(\theta_1|\mathbf{s}_1)\prod_{n=2}^{N}p(\mathbf{s}_n)\cdot p(\theta_n|\mathbf{s}_n,\mathbf{s}_{\phi(n)},\theta_{\phi(n)}) \tag{67}$$

where $\mathbf{s}_{\phi(n)}$ and $\theta_{\phi(n)}$ are the location and direction of minutia $\mathbf{x}_i$, respectively, which is the nearest neighbour to the minutia $\mathbf{x}_n$, with

$$\phi(n)=\arg\min_{i\in[1,n-1]}\|\mathbf{x}_n-\mathbf{x}_i\|, \tag{68}$$

while the conditional density functions used is Eq. (39) along with

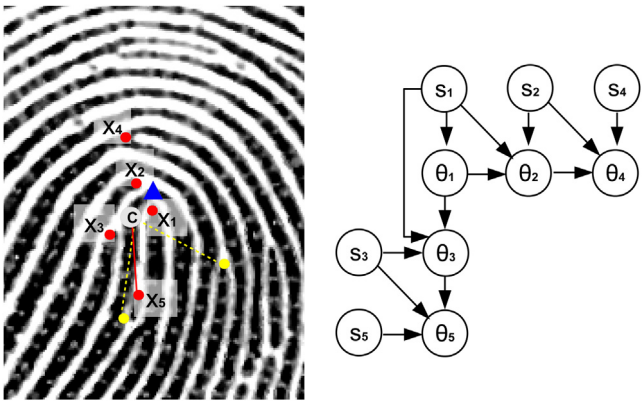


Fig. 3. Left: Minutiae sequencing initial procedure starting off with locating the closest minutia to the core point (as depicted by the blue triangle). The resulting spatial arithmetic mean found after the fifth iteration is depicted as 'c'. Each iteration involves selecting the next minutia of the sequence (being that which is closest to 'c') followed by the recalculation of 'c'. Right: The resulting Bayesian network representing the minutia dependency sequence.

$$f(\mathbf{s}_n) = \sum_{k_1=1}^{K_1} \pi_{k_1} \mathcal{N}(\mathbf{s}_n | \mu_{k_1}, \Sigma_{k_1}) \tag{69}$$

$$f(\theta_n | \mathbf{s}_n, \mathbf{s}_{\phi(n)}, \theta_{\phi(n)}) = \sum_{k_2=1}^{K_2} \pi_{k_2} \cdot V(\theta_n, \nu_{k_2}, \kappa_{k_2}), \tag{70}$$

where K_i is the number of mixture components, π_{k_i} are the component weights, μ_{k_i} and Σ_{k_i} are the mean and covariance matrices, respectively, for the ith group bivariate Gaussian distribution parameters, ν_{k_i} and κ_{k_i} are the mean and precision, respectively, for the ith group Von-Mises distribution parameters, and $\mathbf{s}_i$ along with θ define the input minutia spatial and direction values. The Bayesian Information Criteria (BIC) was used to estimate K_i while the EM algorithm was used to estimate other parameters.

A modified *nPRC* calculation based on the method in [53] was proposed, with additional consideration of a minutia confidence measure, in order to account for varying quality of fingermarks. A confidence measure for minutia $\mathbf{x}_n$ of (d_{s_n}, d_{θ_n}) for the location and orientation was also defined, in which respective confidence distributions were defined as

$$c(\mathbf{s}' | \mathbf{s}_n, d_{s_n}) = \mathcal{N}(\mathbf{s}' | \mathbf{s}_n, d_{s_n}^{-1}) \tag{71}$$

$$c(\theta' | \theta_n, d_{\theta_n}) = V(\theta' | \theta_n, d_{\theta_n}) \tag{72}$$

for a potential matching minutia, $\mathbf{x}' = (\mathbf{s}', \theta')$. The probability that there is a 1-to-1 correspondence between minutia set $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and $\mathbf{X}' = \{\mathbf{x}'_1, \ldots, \mathbf{x}'_{m'}\}$ was given by

$$p_\epsilon(\mathbf{X}) = p_\epsilon(\mathbf{s}_1, \theta_1) \prod_{n=2}^{m} p_\epsilon(\mathbf{s}_n) \cdot p_\epsilon(\theta_n | \mathbf{s}_n, \mathbf{s}_{\phi_n}, \theta_{\phi_n}) \tag{73}$$

where

$$p_\epsilon(\mathbf{s}_n, \theta_n) = \int_{\mathbf{s}'} \int_{\theta'} \iint_{|\mathbf{x}-\mathbf{x}'| < \epsilon} c(\mathbf{s}' | \mathbf{s}_n, d_{s_n}) \cdot c(\theta' | \theta_n, d_{\theta_n}) f(\mathbf{s}, \theta) d\mathbf{s}' d\theta' d\mathbf{s} d\theta, \tag{74}$$

$$p_\epsilon(\mathbf{s}_n) = \int_{\mathbf{s}'} \int_{|\mathbf{s}-\mathbf{s}'| < \epsilon_{\mathbf{s}}} c(\mathbf{s}' | \mathbf{s}_n, d_{s_n}) \cdot f(\mathbf{s}) d\mathbf{s}' d\mathbf{s}, \tag{75}$$

and

$$p_\epsilon(\theta_n | \mathbf{s}_n, \mathbf{s}_{\phi_n}, \theta_{\phi_n}) = \int_{\mathbf{s}'} \int_{|\theta-\theta'| < \epsilon_\theta} c(\theta' | \theta_n, d_{\theta_n}) \cdot f(\theta_n | \mathbf{s}_n, \mathbf{s}_{\phi_n}, \theta_{\phi_n}) d\theta' d\theta. \tag{76}$$

Using the probability of matching exactly m minutiae as

$$p_{f_\epsilon}(m) = \sum_{m_1 \in M} p_c(m_1) \binom{m_1}{m} \cdot \sum_{i=1}^{\binom{m_f}{m}} p_\epsilon(\mathbf{X}'_i), \tag{77}$$

the *nPRC* can simply be calculated with Eq. (63).

3.4.2. Model analysis

This model is more sophisticated than previous generative based models as it includes some representation of inter-minutiae dependency. The inter-minutiae directional dependency was not extended to k-nearest neighbours. However, extending the directional dependency to the k-nearest neighbours may not necessarily be beneficial and will need further investigation. One modelling consideration missing (as with all PRC models) is the incorporation of a distortion model, particularly for *nPRC* calculations. This will improve the accuracy for model PRC calculations. Lastly, the assessment of the model fit solely relied on the Chi-square goodness-of-fit which has the shortcomings discussed earlier in Section 3.3.5.

3.5. Inhomogeneous spatial point process based models

In general, point patterns can be categorised as either over-dispersed, random, or clustered (Fig. 4). Section 3.2 described two models which assume minutia patterns to be a homogeneous distributions of points in order to simplify the model formulation. However, previous studies [59] have observed minutiae patterns to over-disperse on a small scale, while clustering occurs (specifically for core and delta regions) for larger scaled spatial analysis. Thus, for a more accurate representation of minutiae spatial information, a more detailed point pattern model must be formulated using spatial point processes that go beyond the homogeneity assumptions.

3.5.1. Chen et al. [60]

The authors of [60] focused solely on modelling the spatial characteristics of minutiae. A Markov point process was used to generate synthetic minutiae (spatial detail only) with a tendency for over-dispersion on a small scale, while a *thinning process* is used to remove individual synthetic minutiae in order to mimic the large scale clustering tendencies of minutiae patterns. The dispersion/clustering tendencies of a spatial point process can be analysed by the *K function*:

$$K(t) = \lambda^{-1} \cdot E[N(t)] \tag{78}$$

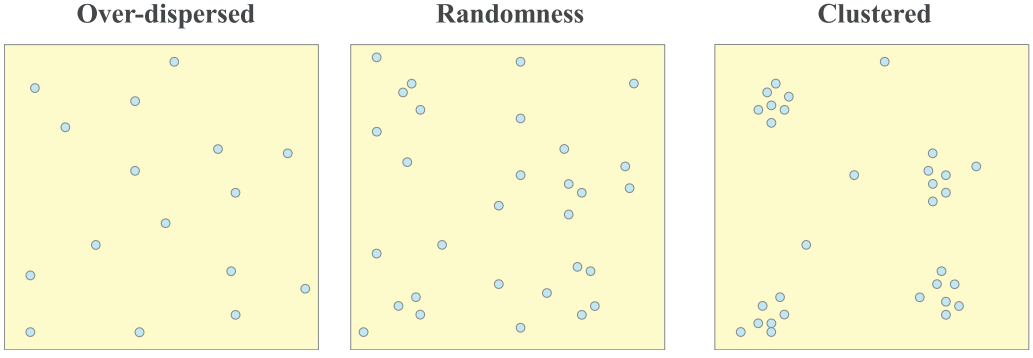


Fig. 4. Left: An over-dispersed (or uniform, regular) point pattern. Centre: A random (i.e., CSR) point pattern. Right: A clustered point pattern.

where $E[N(t)]$ is the expected number of spatial events (i.e., minutiae) to occur within distance t of an arbitrary minutia event. CSR (complete spatial randomness) patterns (Section 3.2.2) which have no clustering or dispersion tendencies converge to a K function value of $K(t) = \pi \cdot t^2$. However, for a finite minutiae pattern, $X = \{x_1, x_2, \ldots, x_n\}$, the K function has to be estimated using Ripley's K function:

$$\hat{K}^*(t) = \frac{|A|}{n(n-1)} \sum_{i=1}^{n} \sum_{j \neq i} \omega(x_i, r_{ij})^{-1} \cdot I(r_{ij} < t) \tag{79}$$

where all n minutiae occur within a planar region A (with area $|A|$), $I(\bullet)$ is the indicator function, r_{ij} is the distance between x_i and x_j, and $\omega(\bullet)$ is a weight function which is the proportion of the circle, centred at x_i with radius r_{ij}, that lies within A.

The authors made an assumption that the clustering and dispersion properties of minutiae patterns can be generalised for an entire population of fingerprints. A Markov point process probability density function is defined as

$$f(X) = \frac{1}{Z} \exp(-U(X)) \tag{80}$$

where Z is a normalisation constant,

$$U(X) = \sum_{i=1}^{n} \sum_{j=i+1}^{n} h(\|x_i - x_j\|), \tag{81}$$

and

$$h(r) = \begin{cases} 0, & \text{if } r < r_0, \\ 1 - \exp((r - r_0) \times (ar + b)), & \text{if } r_0 \leq r < r_1, \\ 1 & \text{if } r_1 \leq r. \end{cases} \tag{82}$$

The pair potential function, $h(\bullet)$, describes attraction and repulsion forces depending on inter-minutiae distances. The weighted average of Ripley's K function estimated per minutiae configuration:

$$\hat{K}^*(t) = \frac{\sum_{i=1}^{N} n_i \cdot \hat{K}_i(t)}{\sum_{i=1}^{N} n_i} \tag{83}$$

was used to discover the parameter values for r_0 and r_1 for a given dataset of fingerprints. The parameter r_0 was set as the smallest inter-minutiae neighbour distance:

$$r_0 = \min_t(\hat{K}^*(t) > 0), \tag{84}$$

whereas r_1 was set as the minimum distance where a clustering tendencies starts:

$$r_1 = \min_t(\hat{K}^*(t) - \pi \cdot t^2 > 0). \tag{85}$$

The thinning process, $Z(x)$, which is another independent stochastic process representing minutiae clustering tendencies, is applied to the synthetic minutiae generated by the Markov point process, acting as a filtration procedure. This involves each generated minutia, x_i, having a probability $Z(x_i)$, to be retained. The authors defined $Z(x)$ as the union of r_δ-radius discs, each centred on a set of points, $Y = \{y_1, y_2, \ldots, y_m\}$ (with $m < n$), which are generated by a Poisson process with intensity λ. Hence,

$$Z(x) = \begin{cases} p & \text{if } x \in \bigcup_{y_i \in Y} R_{y_i}, \\ 1 & \text{otherwise} \end{cases} \tag{86}$$

for some p with $0 < p \leq 1$. Large scale clustering is achieved in areas outside of $Z(x)$, since such in areas are not thinned by $Z(x)$.

Combining the pair potential Markov and thinning processes, there are five unknown parameters of $\zeta = \{a, b, r_\delta, \lambda, p\}$. These can be estimated by using a non-linear minimising algorithm on

$$Q(\zeta) = \int_0^{t_0} \left[(\hat{K}^*(t))^{1/2} - (\hat{K}_S(t;\zeta))^{1/2} \right]^2 dt \tag{87}$$

where $\hat{K}_S(t;\zeta)$ is the point-wise mean of the estimated K functions calculated from S simulated realisations of the proposed point process, while t_0 can be set as the largest inter-minutiae distance.

An experiment was set up where the proposed model was compared to the CSR model (Section 3.2.2) and empirical distributions for the number of matching minutiae for impostor comparisons. Synthetic minutiae were generated from both models, while detected minutiae from three fingerprint databases were used for the empirical distribution. The Hungarian algorithm was used to find optimal pairings of minutiae for each the impostor comparisons. Results indicate that the proposed model's distribution was close to the real world empirical distribution of minutiae spatial detail.

3.5.2. Lim et al. [61]

The authors of [61] proposed a spatial model based on a flexible class of marked point processes and a fully Bayesian inferential framework, where unlike the point process used in [60], each point can be marked with an additional random variable (representing direction). This meant that both minutiae spatial and directional information (including inter minutiae directional dependencies) could be modelled by such a scheme. Inference of model parameters was carried out using a Bayesian MCMC framework. In addition, the EPIC metric was used to compare minutia pattern models.

Given a minutiae pattern, $X = \{x_1, x_2, \ldots, x_n\}$, with respective marks, $W = \{w_{x_1}, w_{x_2}, \ldots, w_{x_n}\}$, a hierarchical model for (x_i, w_{x_i}) was defined as

$$(\boldsymbol{\theta}, \mathbf{m}_{\boldsymbol{\theta}}) = \bigcup_{k=1}^{K} (\theta_k, m_{\theta_k}) \equiv \Phi \sim \mathcal{P}(\lambda_1, h_1) \tag{88}$$

$$(\mathbf{x}^{(k)}, \mathbf{w}_{x^{(k)}}^{(k)}) | \Phi \sim \mathcal{P}(\lambda_{2k}, g_k) \tag{89}$$

for $k = 1, \ldots, K$, and

$$(\mathbf{x}_n, \mathbf{w}_n) = \bigcup_{k=1}^{K} (\mathbf{x}^{(k)}, \mathbf{w}_{x^{(k)}}^{(k)}) \tag{90}$$

where:

- $(\boldsymbol{\theta}, \mathbf{m}_{\boldsymbol{\theta}})$ is a set of K points from the marked Poisson process $\mathcal{P}(\lambda_1, h_1)$ with intensity measure

$$\lambda_1(s) = \begin{cases} K_0 / area(S) & \text{if } s \in S, \\ 0 & \text{otherwise} \end{cases} \tag{91}$$

for some known integer K_0 and rectangular region $S \subset \mathbb{R}^2$, while the joint density function of the marks, h_1, is the compound random variable

$h_1 \sim (\gamma, \sigma_1^2, \sigma_2^2, \eta, \rho, \delta^2)$ with $\gamma \sim G(\alpha_\gamma, \beta_\gamma)$ (i.e., Gamma distribution with shape and scale parameter),

$\sigma_1^2 \sim \mathrm{IG}(\alpha_1, \beta_1)$ (i.e., Inverse Gamma distribution with shape and scale parameter), $\sigma_2^2 \sim \mathrm{IG}(\alpha_2, \beta_2), \eta \sim \mathrm{U}(0, \pi)$ (i.e., Uniform distribution defined on [0, π)), $\rho \sim \mathrm{U}(\rho_{min}, \rho_{max})$, and $\delta \sim \mathrm{G}(\alpha_\delta, \beta_\delta)$. Thus, the point θ_k has a corresponding mark

$m_{\theta_k} = (\gamma_k, {\sigma_{1k}}^2, {\sigma_{2k}}^2, \eta_k, \rho_k, {\delta_k}^2)$. These distributions are parameters for the density function defined in the next upward level. All hyper-parameters for the Gamma and Inverse Gamma prior distributions are assumed to be known and fixed.

- The intensity measure in Eq. (89) is

$$\lambda_{2k}(s) = \begin{cases} \gamma_k . \mathcal{N}_2(s|\theta_k, \sigma_{1k}^2, \sigma_{2k}^2) & \text{if } s \in S, \\ 0 & \text{otherwise} \end{cases} \tag{92}$$

where $\mathcal{N}_2(s|\theta_k, \sigma_{1k}^2, \sigma_{2k}^2)$ is a bivariate Normal distribution which the models spatial density of the kth cluster. The corresponding joint density function of the marks is

$$g_k(\mathbf{w}|\eta_k, \rho_k, \delta_k^2) = \mathbf{V}_{n_k}\left(\mathbf{w}|\eta_k, \sum_k(\rho_k, \delta_k^2)\right) \tag{93}$$

where $\mathbf{V}_{n_k}$ is the n_k -variate wrapped normal distribution on [0, π) with mean $\eta_k = (\eta, \ldots, \eta)' \in \mathbb{R}^{n_k}$ and covariance matrix $\sum_k(\rho_k, \delta_k^2) = (\sigma_{rs}^*)$ with $r, s = 1, 2, \ldots, n_k$ with entries

$$\sigma_{rs}^* = \delta^2 \exp(-\rho \|x_r - x_s\|), \tag{94}$$

where $x_r, x_s \in \mathbf{x}^{(k)}$. The wrapped normal distribution is used to model orientations for the n_k points in the kth group.

- The marked points ($\mathbf{x}_n$, $\mathbf{w}_n$) is simple the union of the marked points from each of the K groups of spatial/orientation marked points.
- The number of groups, K, along with each parameter in groups are updated using a Markov Chain Monte Carlo based algorithm with a minimum and maximum set to $K_{min} = 2$ and $K_{max} = 5$, respectively. The DIC (Deviance Information Criteria) was used to assess the model fit.

After each fingerprint feature models completed simulation (i.e., when convergence of parameters occurred), inference based on EPIC (see Section 2.1) was performed using N_0 samples generated from each fitted marked point process. Each sample from the jth feature model consists of M_j^l synthetic minutiae. For two model samples (i.e., $j = 1, 2$) a pairing of minutiae (M_1^l, M_2^l) was performed for all $l = 1, \ldots, N_0$ samples. The number of synthetic minutiae pairs for each sample comparison was used to construct an estimate for the probability distribution, $\mathcal{S}$, for the number of impostor minutiae pairs. While denoting the empirical number of matching minutiae, τ_0, from the two original representative fingerprint configurations of both models, an estimate for EPIC was calculated as

$$EPIC(\tau_0) = P(\mathcal{S} \geq \tau_0). \tag{95}$$

3.5.3. Other related models

One other model known to use an inhomogeneous spatial representation of minutiae spatial detail can be found in [62,63]. In [62], Poisson point processes were used for different regions of the fingerprint, where each region had an estimated intensity parameter, λ. For instance, regions centred at singularities such as cores and deltas had a high value λ in comparison to periphery regions. In [63], a hierarchical mixture model is created with the top level representing homogeneous groups of fingerprints in a given population and a second level made up of nested mixtures are used to flexibly represent the distributions of minutiae. A Bayesian approach using MCMC framework similar to [61] was used to find all unknown parameters of the hierarchical mixtures.

3.5.4. Model methodology analysis

Inhomogeneous point process models have the potential to accurately replicate minutiae pattern characteristics of different scales. The model in [60] accurately represents small scale over-dispersion and large scale clustering of a given population. However, the authors make an unfounded assumption that each fingerprint has largely similar clustering and over-dispersion characteristics.

The model in [61] differs by creating a feature model per fingerprint. This allows the model to accurately model spatial clustering tendencies along with directional dependencies between neighbouring minutiae. However, the authors noted that the feature models are slow to converge. In addition, no extensive goodness-of-fit tests were performed on the resulting feature models.

4. Likelihood Ratio based models

Likelihood Ratio (LR) models can be classified as Feature Vector (FV) or AFIS score based. Both methods rely on distributions of analytically based metrics for within-finger and between-finger populations to derive an LR value.

4.1. Feature Vector based LR models

FV based LR models are based on FVs containing various minutiae feature analyses. A dissimilarity metric is defined on the constructed FVs, from which the distributions of the defined dissimilarity metrics on both within-finger and between-finger comparisons are used to calculate an LR value.

4.1.1. Neumann et al. [38]

The first attempt to create a FV based LR model was proposed in [38], where FVs were constructed from the Delaunay triangulation (Fig. 5, left) of minutiae. Each FV was constructed as follows:

$$x = \{GP_x, R_x, Nt_x, (A_{1x}, L_{1x-2x}), (A_{2x}, L_{2x-3x}), (A_{3x}, L_{3x-1x})\} \tag{96}$$

where GP_x is the pattern of the mark, R_x is the region of the fingerprint, Nt_x is the number of minutiae that are ridge endings constituting the triangle (with $Nt_x \in \{0, 1, 2, 3\}$), A_{ix} is the angle of the ith minutia, and $L_{ix-((i+1)\mathrm{mod}3)x}$ is the length in pixels between the ith and the $((i+1)\mathrm{mod}3)$th minutiae. Likewise, these structures are created for candidate fingerprint configurations:

$$y = \{GP_y, R_y, Nt_y, (A_{1y}, L_{1y-2y}), (A_{2y}, L_{2y-3y}), (A_{3y}, L_{3y-1y})\}. \tag{97}$$

The FVs can be decomposed into *continuous* and *discrete* components, representing the measurement based on continuous and count/categorical features, respectively. The proposed LR is given as:

$$LR = \underbrace{\frac{P(x_c, y_c|x_d, y_d, H_P)}{P(x_c, y_c|x_d, y_d, H_D)}}_{LR_{c|d}} \cdot \underbrace{\frac{P(x_d, y_d|H_P)}{P(x_d, y_d|H_D)}}_{LR_d} = LR_{c|d}.LR_d \tag{98}$$

where LR_d is the Likelihood Ratio dealing with discrete FVs $x_d = \{GP_x, R_x, Nt_x\}$ and $y_d = \{GP_y, R_y, Nt_y\}$, while continuous FVs x_c and

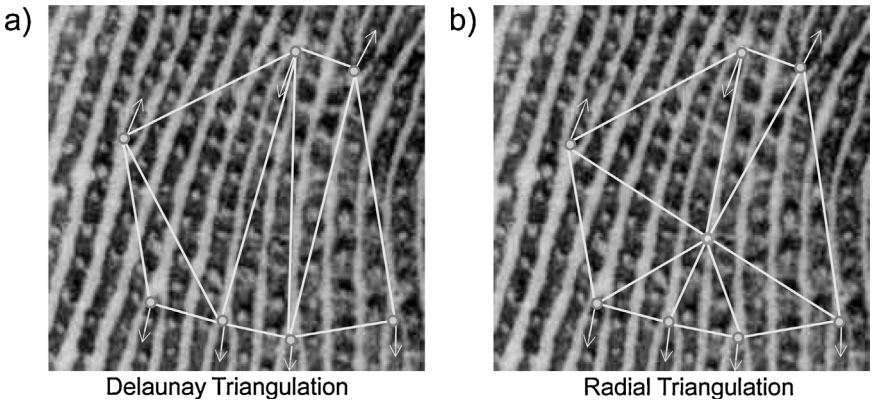


Fig. 5. Delaunay triangulation (left) and radial triangulation (right) differences for a configuration of 7 minutiae. The blue point for the radial triangulation illustration represents the centroid (i.e., arithmetic mean of minutiae x–y coordinates).

y_c contain then remaining features in x and y, respectively. The discrete likelihood numerator takes the value of 1, while the denominator was calculated using frequencies for general patterns multiplied by region and minutia-type combination probabilities observed from large datasets.

A dissimilarity metric, $d(x_c, y_c)$, was created for comparing the continuous FV defined as:

$$d(x_c,y_c) = \Delta^2 A_1 + \Delta^2 L_{1-2} + \Delta^2 A_2 + \Delta^2 L_{2-3} + \Delta^2 A_3 + \Delta^2 L_{3-1} \quad (99)$$

with Δ^2 as the squared difference of corresponding variables from x_c and y_c. This was used to calculate the continuous likelihood value, with:

$$LR_{c|d} = \frac{P(d(x_c,y_c)|x_d,y_d,H_P)}{P(d(x_c,y_c)|x_d,y_d,H_D)}. \quad (100)$$

Density functions of both the numerator and denominator of Eq. (100) were estimated using a kernel smoothing method. All $LR_{c|d}$ numerator and denominator likelihood calculations were derived from these distribution estimates.

Two experiments were configured in order to evaluate within-finger and between-finger LRs. Ideally, LRs for within-finger comparisons should be larger than all between-finger ratios. The within-finger experiment used 216 fingerprints from 4 different fingers under various different distortion levels. The between-finger datasets included the same 818 fingerprints used in the minutia-type probability calculations. The Delaunay triangulation had to be manually adjusted in some within-finger cases due to different triangulation results occurring under high distortion levels. Error rates for LRs greater than 1 for between-finger and LRs less than 1 for within-finger comparisons for index, middle, and thumbs, are given in Table 3. These error rates indicate the power that the proposed FVs have in distinguishing within and between finger configurations.

Table 3
Some likelihood Ratio error rate results for different finger/region combinations.

Finger	Region	LR true <1	LR false >1
Index	All	2.94%	1.99%
Index	Core	4.19%	1.36%
Index	Delta	1.95%	2.62%
Middle	All	1.99%	1.84%
Middle	Core	3.65%	1.37%
Middle	Delta	2.96%	2.58%
Thumb	All	3.27%	3.24%
Thumb	Core	3.74%	2.43%
Thumb	Delta	2.39%	5.20%

4.1.2. Neumann et al. [38]

While the proposed triangular structures of [38] produced an accurate dichotomy between within-finger and between-finger comparisons, there was an observed issue with the proposed FV structure's robustness with some cases of distortion. In addition, the LR model could make more accurate assessments by including more minutiae (i.e., more information) in the FV structures, rather than restricting each FV to only have three minutiae.

The authors of [39] defined *radial triangulation* FVs based on n minutiae $x = \{GP_x, x^{(n)}\}$ where GP denotes the general pattern and

$$x^{(n)} = \{(\delta_i, \sigma_i, \theta_i, \alpha_i, \tau_i): \quad i = 0, 1, \ldots, n-1\},$$

is Feature Vector for a set of minutiae numbered in a clockwise order of $i = 0, 1, \ldots, n-1$, where δ_i is the distance between the ith minutia and the centroid point (Fig. 5), σ_j is the distance between the ith minutia and the next contiguous minutia (in a clockwise direction), θ_j is the angle between the direction of a minutia and the line from the centroid point, α_i is the area of the triangle constituted by the ith minutia, the next contiguous minutia and the centre of the polygon, and τ_i is the type of the ith minutia (i.e., ridge ending or bifurcation).

The LR was calculated as

$$LR = \underbrace{\frac{P(x^{(n)},y^{(n)}|GP_x,GP_y,H_P)}{P(x^{(n)},y^{(n)}|GP_x,GP_y,H_D)}}_{LR_{n|g}} \times \underbrace{\frac{P(GP_x,GP_y|H_P)}{P(GP_x,GP_y|H_D)}}_{LR_g} = LR_{n|g} \times LR_g \quad (101)$$

where LR_g was formed with $P(GP_x, GP_y|H_P) = 1$ and $P(GP_x, GP_y|H_D)$ equal to the FBI pattern frequency data. Noting that the centroid FVs can be arranged in n different ways (accounting for clockwise rotation):

$$y_j^{(n)} = \{(\delta'_k, \sigma'_k, \theta'_k, \alpha'_k, \tau'_k): \; k = j, (j+1) \bmod n, \ldots, (j-1) \bmod n\},$$

for $j = 0, 1, \ldots, n-1$, $LR_{n|g}$ was defined as

$$LR_{n|g} = \frac{P(d(x^{(n)},y^{(n)})|GP_x,GP_y,H_P)}{P(d(x^{(n)},y^{(n)})|GP_x,GP_y,H_D)} \quad (102)$$

with the defined dissimilarity metric

$$d(x^{(n)}, y^{(n)}) = \min_{j=0,\ldots,n-1} d(x^{(n)}, y_j^{(n)}). \quad (103)$$

Each $d(x^{(n)}, y_j^{(n)})$ was calculated as the Euclidean distance of respective FVs that have been normalised to take a similar range of values. The two conditional probability densities of Eq. (102) were estimated using mixture models of normal distributions with a mixture of three and four distributions, respectfully, using the EM algorithm to estimate distributions for each finger/configuration size used.

This method modelled within and between finger variability both accurately and robustly, due to the flexibility of the centroid structures containing more than three minutiae. For example, the addition of one extra minutia halved the LR error rate for some fingerprint patterns. In addition, the prior LR in Eq. (101) is more flexible in comparison to the discrete LR component of Eq. (98) for real life applications, as it is not dependent on identifying the specific fingerprint region (which is robust for real life fingermark-to-exemplar comparisons).

4.1.3. Neumann et al. [40]

Another FV based LR model using radial triangulation structures was proposed in [40], where the model was further refined to use distortion and examination influence models. The radial triangulation FVs used were based on the structures defined in [39], with the minutiae type feature, τ, extended to account for defined types of ridge ending, bifurcation, and unknown.

The distance between configurations $x^{(n)}$ and $y^{(n)}$, each representing n minutiae, with $y_i^{(n)}$ denoting the ith cyclic rotation of $y^{(n)}$, was defined as

$$d(x^{(n)}, y^{(n)}) = \min_{i=0,\ldots,n-1} d_c(x^{(n)}, y_i^{(n)}) \quad (104)$$

$$d_c(x^{(n)}, y_i^{(n)}) = \sum_{j=1}^{n} \Delta_j \quad (105)$$

$$\Delta_j = q_\delta \cdot (\delta_j - \delta'_j)^2 + q_\sigma \cdot (\sigma_j - \sigma'_j)^2 + q_\theta \cdot d_\theta(\theta_j, \theta'_j)^2 + q_\alpha \cdot (\alpha_j - \alpha'_j)^2 + q_\tau \cdot d_T(\tau_j, \tau'_j)^2 \quad (106)$$

where using normalised values for all features, d_θ is the angular difference and d_T is the defined minutiae type difference metric. The multipliers (i.e., q_δ, q_σ, q_θ, q_α, and q_τ) are tuned via a heuristic based procedure.

The proposed LR model makes use of a distortion model based on the Thin Plate Spline (TPS) bending energy matrices representing the non-affine differences of minutiae spatial detail trained from a dataset focused on finger variability, and an examiner influence model created to represent the variability of examiner markings of minutiae in fingerprint images. Given $y^{(k)}$ as a configuration within a fingermark, $x_{\min}^{(k)}$ as the closest k configuration found, and $z_{i,\min}^{(k)}$ as the closest configuration for the ith member of a reference database containing N impressions, *synthetic* FVs can be generated from small scale modifications to minutiae locations, via Monte-Carlo simulation of both distortion and examiner influence models. A set of M synthetic FVs are created for $x_{\min}^{(k)}$ ($\{\zeta_1^{(k)}, \ldots, \zeta_M^{(k)}\}$) and for each $z_{i,\min}^{(k)}$ ($\{\zeta_{i,1}^{(k)}, \ldots, \zeta_{i,M}^{(k)}\}$), from which the LR is calculated as

$$LR = \frac{N \sum_{i=1}^{M} \psi(d(y^{(k)}, \zeta_i^{(k)}))}{\sum_{i=1}^{N} \sum_{j=1}^{M} \psi(d(y^{(k)}, \zeta_{i,j}^{(k)}))}. \quad (107)$$

The transformation function ψ on the dissimilarity metric values is defined as

$$\psi(d(y^{(k)}, \bullet)) = \exp\left(\frac{-\lambda_1 d(y^{(k)}, \bullet)}{T^{(k)}}\right) + \frac{B(d(y^{(k)}, \bullet), \lambda_2 k)}{B(d_0, \lambda_2 k)} \quad (108)$$

which is a mixture of Exponential and Beta functions tuned with parameters λ_1 and λ_2, to meet the following *desiderata*:

- the distribution of $LR(H_P)$ will predominantly have large values greater than 1,
- the distribution of $LR(H_D)$ will predominantly have large values less than 1, and
- minimal variation of LR values resulting from markings made by different examiners on a given fingermark-to-template comparison,

while d_0 is the smallest value into which distances were binned, and $T^{(k)}$ is the 95th percentile of simulated scores from the examiner influence model applied on $y^{(k)}$.

Experimental results from a validation dataset illustrated that the proposed LR model can generally distinguish within and between finger comparisons with a high degree of accuracy, while an increased dichotomy arose from increasing the configuration size. In addition, the closest non-match candidates returned from an AFIS containing 600 million fingerprints were also evaluated by the model with high accuracy.

4.1.4. Abraham et al. [64]

The LR model found in [64] uses FVs constructed from shape descriptors calculated on pre-aligned corresponding minutiae configuration samples from both match (i.e., within-finger) and close non-match (i.e., similar between-finger) populations. Given two equally sized corresponding and pre-aligned minutiae configurations, $x^{(m)}$ and $y^{(m')}$, with x–y components denoted as $\mathbf{X}$ and $\mathbf{Y}$, respectively, the shape descriptors used to construct the FV, $\mathbf{x}_i$, include the Euclidean Distance Matric Analysis (EDMA) test statistic (T), three dimensional Kolmogorov–Smirnov statistic ($Z_{x^{(m)}, y^{(m')}}$), centroid shape size ($S(\mathbf{X})$), ordinary partial Procrustes sum of squares ($OSS_p(\mathbf{X_c}, \mathbf{Y_c})$ with centred x–y coordinate configurations $\mathbf{X_c}$ and $\mathbf{Y_c}$), and Thin Plate Spline (TPS) based metrics describing affine (d_θ, d_{shear}, d_{scale}, d_{offset}) and non-affine (bending energy, I_f) characteristics of the aligned minutiae configurations, giving

$$\mathbf{x}_i = \{T, Z_{x^{(m)}, y^{(m')}}, S(\mathbf{X}), d_S, OSS_p(\mathbf{X_c}, \mathbf{Y_c}), I_f, d_\theta, d_{shear}, d_{scale}, d_{offset}\}.$$

Such shape descriptors are proposed in order to separate "valid" and "invalid" skin distortion characteristics of match and close non-match cases, respectively.

Corresponding minutiae configurations were derived from candidate lists of AFIS searches using a set of alignment algorithm and score thresholds, K, empirically set to accept all within-finger configurations of a test set. Using such candidate lists, Eq. (3) can be re-written as

$$LR = LR_K \times \frac{P(K|y^{(m')}, H_P)}{P(K|y^{(m')}, H_D)} \quad (109)$$

where LR_K is the candidate list LR defined as

$$LR_K = \frac{P(x^{(m)}|y^{(m')}, K, H_P)}{P(x^{(m)}|y^{(m')}, K, H_D)}, \quad (110)$$

while $P(K|y^{(m')}, H_P)$ and $P(K|y^{(m')}, H_D)$ are the probabilities that for configuration, $y^{(m')}$, match and non-match configuration samples,

respectively, are in a candidate list containing all candidates meeting K. For an ideal scenario where the probability $P(K|y^{(m')}, H_P)$ is close to 1, an estimate for LR is given as

$$LR \approx \frac{LR_K}{P(K|y^{(m')}, H_D)}. \quad (111)$$

The constructed FVs were used to train a commonly used machine learning classifier, called Support Vector Machines (SVMs), in an attempt to learn the shape descriptor relationships for minutiae correspondences from match and close non-match comparisons. SVMs were constructed and trained for each configuration size using test sets of FVs categorised by size and comparison class (i.e., match or close non-match). Denoting match and close non-match classes as $\mathbf{M}$ and $\mathbf{M}'$, respectively, LR_K was calculated for an FV, $\mathbf{x}_i$, as

$$LR_K = \frac{P(f(\mathbf{x}_i)|\mathbf{x}_i \text{ is class } \mathbf{M})}{P(f(\mathbf{x}_i)|\mathbf{x}_i \text{ is class } \mathbf{M}')} \quad (112)$$

$$= \left(\frac{1-a}{a}\right) \cdot \frac{P(\mathbf{x}_i \text{ is class } \mathbf{M}|f(\mathbf{x}_i))}{P(\mathbf{x}_i \text{ is class } \mathbf{M}'|f(\mathbf{x}_i))}. \quad (113)$$

where $a = P(\mathbf{x}_i \text{ is class } \mathbf{M})$, A and B are estimated parameters of a fitted logistic link function,

$$P(\mathbf{x}_i \text{ is class } \mathbf{M}|f(\mathbf{x}_i)) = \frac{1}{1+\exp(Af(\mathbf{x}_i)+B)}, \quad (114)$$

and

$$P(\mathbf{x}_i \text{ is class } \mathbf{M}'|f(\mathbf{x}_i)) = 1 - P(\mathbf{x}_i \text{ is class } \mathbf{M}|f(\mathbf{x}_i)). \quad (115)$$

An evaluation set containing real life close non-match examples was used to test the accuracy of the model's derived LR value, in terms of how often the LR value favoured the correct hypothesis. Experimental results revealed that the proposed model is highly accurate in favouring the correct hypothesis when assessing close non-match and match configuration comparisons. For example, configurations with 5 minutiae had a Rate of Misleading Evidence in favour of Defence (RMED) of 9.9% and Rate of Misleading Evidence in favour of Prosecution (RMEP) of 11.4% for candidate list entries, while the increase of configuration size correlated with significant decreases in these rates (e.g., configurations with 8 minutiae had an RMED and RMEP of 3.4% and 3.2%, respectively).

4.1.5. Model methodology analysis

The LR models proposed in [38,39] use dissimilarity measures of FVs which are potentially not robust to real world scenarios, as minutiae types can change, particularly in distorted impressions. While the method in [40] has clearly improved the dissimilarity function by introducing tuned multipliers, squared differences in angle, area, and distance based measures are ultimately not probabilistically based. A joint probabilistic based metric for each FV component using distributions for both impostor and genuine populations would be more consistent with the overall LR framework. A commentator in [40] pointed this out, noting that the LR calculated is not of the traditional Bayesian understanding of an LR (i.e., used to multiply a prior probability in order to get a posterior probability). Nonetheless, the derived LR is still a useful metric for evaluating the strength of evidence for corresponding minutiae configurations.

The radial triangulation FV structures of [39,40] are robust towards skin distortion, unlike the Delaunay triangulation structure of [38]. Furthermore, the model proposed in [40] models realistic skin distortion encountered on flat surfaces by measuring the bending energy matrix for a specialised distortion set. However, this only accounts for the non-affine variation. Affine transformations such as shear and uniform compression/dilation are not accounted for. Such information can be particularly significant for comparisons of small minutiae configurations encountered in fingermarks. For instance, a direct downward application of force may have prominent shear and scale variations (in addition to non-affine differences) for minutiae configurations, in comparison to the corresponding configurations of another impression from the same finger having no notable downward force applied. Also, the LR calculation is computationally expensive as it requires a direct comparison with the entire reference set for every new evaluation.

The model described in [64] has a practical advantage over other FV methods. This is because FVs can be immediately constructed for a given candidate list returned from an AFIS search, and used as input vectors to trained SVMs, in order to calculate per candidate LR values. In contrast, other models either require FVs comparisons or metric distributions constructed from all configurations in a reference set. In addition, both affine and non-affine characteristics of "valid" skin distortion are accounted for in the FV model. However, there is no consideration of contextual bias in this model.

4.2. AFIS score based LR models

AFIS score based LR models use estimates of the genuine (i.e., within-finger) and impostor (i.e., between finger) similarity score distributions from fingerprint matching algorithm(s) within AFIS, in order to derive an LR.

4.2.1. Egli et al. [41]

In order to estimate the score distributions used in Eq. (5), the authors of [41] proposed using the Weibull $W(\lambda, \beta)$ and Log-Normal $\ln\mathcal{N}(\mu, \sigma^2)$ distributions with scale/shape parameters tuned to estimate the genuine and impostor AFIS score distributions, respectively. Given query and template fingermarks with an AFIS similarity score, s, the LR is

$$LR = \frac{f_W(s|\lambda, \beta)}{f_{\ln\mathcal{N}}(s|\mu, \sigma^2)} \quad (116)$$

using the proposed probability density functions of the estimated AFIS genuine and impostor score distributions. An updated variant can be found in [42], where impostor and genuine score distributions are modelled per minutiae configuration. This allows the rarity of the configuration to be accounted for. This approach was extensively tested in [42]. It is shown that the estimation of the numerator likelihood require a reasonable amount of prints from the source of interest (ideally 8 impressions). It also highlighted that the denominator likelihoods are highly dependant on the number of minutiae, finger and general pattern. Hence the probability densities of interest have to be recomputed on a case by case basis.

4.2.2. Choi et al. [43]

The authors of [43] proposed a model based on AFIS score distributions, using the NMP of Eq. (13). Three main methods for modelling the AFIS score distributions where tested, being (i) histogram based, (ii) Gaussian kernel density based, and (iii) parametric density based estimation using the proposed distributions found in [41]. Given an AFIS score, s, the NMP and LR were calculated by setting $P(H_P) = P(H_D)$, while estimating both $P(s|H_P)$ and $P(s|H_D)$ either by normalised bin (method (i)) or probability density (methods (ii) and (iii)) values for respective distributions.

Experimentation revealed that the parametric method was biased. In addition, the authors suggest that the kernel density method is the most ideal, as it does not suffer from bias while it can be used to extrapolate NMP scores where no match has been observed, unlike the histogram based method.

4.2.3. Model methodology analysis

AFIS score based LR models provide a framework that is both practically based and simple to implement in conjunction with the AFIS architecture. AFIS LR models are the closest to a practical implementation, as they are entirely based on the readily available data that every laboratory have access to. This allows such models to be readily tuned to specific practical applications. However, model performance is largely dependent on the matching algorithm of the AFIS. In fact, LR models presented will usually reflect the exact information contained in a candidate list of an AFIS query. A more complex construction, for instance, multiple AFIS matching algorithms with a mixture-of-experts statistical model would be more ideal and avoid LR values that are strictly algorithm dependent.

The scores produced from matching algorithms in AFIS detail pairwise similarity between two impressions (i.e., mark and exemplar). However, the methods used in [41,43], which generalise the distributions for all minutiae configurations, do not allow evidential aspects such as the rarity of a given configuration to be considered. A more sound approach would be to base LR calculations on methods that do not have primary focus on only pairwise similarities, but consider statistical characteristics of features within a given population. For instance, the LR for a rare minutiae configuration should be weighted to reflect its significance. This is achieved in the method described in [42] by focusing distribution estimates of scores for each minutiae configuration. However, a parametric fit is used, which may not provide adequate for all AFIS score distributions.

5. Discussion and conclusion

PRC models are designed to represent statistical characteristics of minutiae spatial and directional detail by constructing feature models, from which random samples are generated and PRC values calculated. The ultimate focus of PRC models is to evaluate the rarity of feature configurations. Overall, modern PRC models provide an important representation of the statistical relationships of fingerprint features. The current state of PRC models can be summarised as follows:

- Spatial homogeneity models provide an estimate of PRC values using simple mathematically derived formulae. In particular, the model in [48] (Section 3.2.2) provides similar results to empirical PRC values. However, more complex and important statistical properties are ignored.
- General minutiae spatial and orientational clustering tendencies are modelled by spatio-directional based models (Section 3.3).
- Intrinsic inter-minutiae dependencies between neighbouring minutiae orientation/direction are modelled in [57] (Section 3.4.1) and [61] (Section 3.5.2).
- Detailed modelling of minutiae spatial patterns is achieved by the inhomogeneous spatial point process models (Section 3.5).
- The model in [57] (Section 3.4.1) makes consideration for quality assessment by including a confidence weighting for individual minutiae location and orientation detail.
- No currently known PRC model explicitly considers important practical aspects of fingerprint identification, such as skin distortion characteristics and such as the variances in marking minutiae location introduced by human fingerprint practitioners.
- Most PRC models lack the application of a sound evaluation framework. Many proposed models either have not reported a thorough evaluation or rely on non-robust goodness-of-fit statistics for evaluating feature model fit.
- While LR models can derive a PRC value, the reverse is not true. An LR cannot be obtained from PRC values because the within-finger correspondence is not considered (i.e., the LR numerator).

LR models provide a more practically based analysis of minutiae configurations, since empirical data is directly used for an evidentially focused analysis. Such analysis goes beyond attempting to quantify the rarity of a fingerprint feature configuration. In addition, LR models may include real world considerations that fingerprint practitioners consider in identification assessments, such as effects of skin distortion and impact of the examiner. The current state of LR models can be summarised as follows:

- The FV based LR models (Section 4.1) are powerful models which focus on the intrinsic spatial detail of a configuration. This somewhat mimics the behaviour of a human expert.
- The AFIS based LR models (Section 4.2) are simple yet effective models from which practical integration with AFIS is straight forward. The AFIS can simply be treated as a black box to which the LR model is built onto.
- The AFIS based LR models (Section 4.2) are dependent on a single AFIS score rating. A mixture-of-experts statistical model would be a better scenario as combining matching algorithms would result in increased accuracy.
- The FV based LR model proposed in [40] (Section 4.1.3) incorporates a model for skin distortion on flat surfaces. However, this model fails to include affine distortion characteristics such as shear, scale, and compression.
- The FV based LR model proposed in [40] (Section 4.1.3) incorporates a model representing the variance in location of marked minutiae encountered by human experts.
- All FV based LR models (see Section 4.1) illustrate a highly accurate demarcation of genuine and impostor comparisons via the prescribed LR calculations.
- Most FV based LR models (see Section 4.1) consider noisy features such as minutia type, which may prove detrimental in practical applications.
- The FV based LR model found in [64] can be easily integrated with AFIS like AFIS based LR models and is independent of any vendor specific AFIS scoring methodology.

It is evident that both LR and PRC models have excelled in sophistication and practicality in recent years and will have a role to play in a very near future to assist fingerprint practitioners. The major evolution from the seminal review papers by Stoney [27,28] is that these models are increasingly assessed against datasets in order to better understand their strengths and limitations. The discipline has moved from a state where theoretical models were proposed without any strong empirical support for the underpinning assumptions to a state where the rates of misleading evidence can be quantified and exposed. For reasons explained before (e.g. allowance for distortion and examiner variation), we favour models leading to an assignment of LR over models computing PRC.

In our opinion, the assistance brought by LR-based models to fingerprint practitioners will take the following forms:

1 As a way to support decisions made in the actual application of ACE-V. The most recent definition of individualization according to SWGFAST reads as [65]: "Individualization of an impression to one source is the decision that the likelihood the impression was

made by another (different) source is so remote that it is considered as a practical impossibility".

In that context, LR models will offer a mechanism to help with the examiner's assignment of the strength of support required to reach such a decision. That will allow practitioners to move from an opinion expressed ipse dixit to a more transparent process where the strength of support brought by part of the features considered toward the identification can be transparently exposed. Cole [2] rightly stated that "[...] there is a sense in which 'opinion' can become an all-encompassing shield that deflects all accountability". The LR models will aim at providing a transparent mechanism of accountability.

In such a setting, the use of the LR model amounts to adding another expert in the ACE-V process (typically either in Evaluation or Verification). The detailed standard operating procedures (including the management of cases of conflicting results) regulating the deployment of such LR tools in concert with traditional fingerprint experts will need to be worked out, but in doing so, this should address some of the major concerns raised by the critical commentators.

2 LR models can also help in the Analysis phase of ACE-V, where fingermarks need to be assessed for suitability. The LR tool can be used to assign a value to the expected weight of support that may be derived from the features at hand before undertaking any comparison. Allocation of resources could be adapted as a function of the magnitude of the expected LR (in addition to other operational and case-specific constraints).

3 LR models offer a mechanism to assign weight of support (in favour of H_P or H_D) to fingermarks that would be otherwise declared to be inconclusive. In these cases, the evaluation could benefit from the guidance given by the LR model and a statement issued expressing the strength of support associated with the features found in agreement or disagreement. That would represent a radical change in the way fingermark evidence has been traditionally reported and as raised by Campbell [66]: "38.84 – Before a finding of 'unable to exclude' is led in evidence, careful consideration will require to be given to (a) the types of mark for which such a finding is meaningful and (b) the proper interpretation of the finding. An examiner led in evidence to support such a finding will require to give a careful explanation of its limitations."

But overall, LR models would allow the fingerprint discipline to move away from an all-or-nothing approach and come on a par with other forensic disciplines, including the provision of information for investigative and intelligence purposes.

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